

SEPARATOR ALGEBRAS AND TRANSFER THEORY FOR THE ZERO-FORCING POLYNOMIAL

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ABSTRACT. For a finite graph G , let

$$\mathcal{Z}(G; x) = \sum_k z(G; k) x^k,$$

where $z(G; k)$ is the number of zero-forcing sets of size k . Boyer et al. asked for splitting formulas for this polynomial over cut vertices and separating sets. We give such a formula for every finite labeled separator.

The construction uses forts. For a separator of order q , a local obstruction is encoded by a join-closed family of words in $\{0, 1, \mathbf{l}\}^q \setminus \{0^q\}$. We give the exact rule for gluing two local families and express the boundary choice as a transversal problem on the resulting fort traces. The corresponding connection kernels admit a Möbius compression indexed by chains. Every chain state is realized by a finite simple graph, so the finite chain kernel has the same rank as the infinite zero-forcing connection matrix.

For an independent boundary this rank is the number c_q of chains in $C_3^q \setminus \{0^q\}$. In particular $c_1 = 4$, $c_2 = 52$, $c_3 = 1636$, and $\log c_q = 2q \log q + O(q)$. With arbitrary boundary edges the first full ranks are $r_0 = 1$, $r_1 = 4$, and $r_2 = 60$, the last value certified by an exact finite calculation. At one terminal the theory reduces to $\mathbb{Q}(x)[s, a]/(s^2 - s, a^2)$; the associated transfers give explicit quadratic coefficient-operation algorithms for trees and block-cactus graphs.

1. INTRODUCTION

Zero forcing starts from a set of blue vertices. A blue vertex may force its unique white neighbour, and an initial set is zero forcing when repeated legal forces colour the whole graph. The parameter arose from minimum-rank and maximum-nullity questions for symmetric matrices described by a graph [1].

Boyer, Brimkov, English, Ferrero, Keller, Kirsch, Phillips, and Reinhart [2] introduced

$$\mathcal{Z}(G; x) = \sum_{k=0}^{|V(G)|} z(G; k) x^k,$$

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where $z(G; k)$ counts zero-forcing sets of size k . They asked, among other things, for splitting formulas based on cut vertices or separating sets. We answer that question for separators of arbitrary finite order.

We use connection matrices to state the answer. Two q -boundaried graphs are glued by identifying vertices with the same boundary label, and the entry of the connection matrix is the zero-forcing polynomial of the glued graph. The rank records how much linear information must pass through the boundary. This language is standard in graph algebras [24, 25]; in particular, general finite-rank and bilinear-reduction theorems are already known for suitable definable graph polynomials. Our point is more specific: we give the zero-forcing interface explicitly, reduce it to a finite kernel, and determine that kernel exactly enough to recover the connection rank.

The static obstruction is a fort. A nonempty set F is a fort if no vertex outside F has exactly one neighbour in F , and a set is zero forcing if and only if it meets every fort [1]. The fort hypergraph and its transversals have been studied in their own right [26]. Recent work also counts inclusion-minimal forts, including the linear-time tree algorithm of Abiad and Ghasemi Nezhad [6, 7, 8, 5]. That is a different counting problem. Here the object being counted is every subset that meets every fort, with its cardinality retained.

Let B be a separator of order q . We impose the fort condition away from B and call the resulting object a prefort. At a boundary vertex only three pieces of information matter: the vertex lies in the prefort, it lies outside and has no interior prefort neighbour, or it lies outside and has exactly one. Counts at least two are already safe. The resulting atoms lie in

$$P_q = \{0, 1, \mathbf{l}\}^q \setminus \{0^q\}.$$

For a fixed interior initial set the atom family is join closed in the product order $0 < 1 < \mathbf{l}$. If two atoms have the same trace T and the glued boundary has edge set E , they form a fort exactly when

$$d_E(i, T) + \alpha_i + \beta_i \neq 1 \quad (i \notin T).$$

The selected boundary vertices must then meet every trace produced by a compatible pair. This is the splitting formula.

The local state space is large, but its linear content is smaller. A Möbius cancellation on finite join-semilattices leaves only chains. We prove that every chain state is realized by a finite simple graph and use one-terminal selection projectors to isolate its prescribed interior set. Hence no mode in the compressed kernel is formal only. If $M_E(x)$ denotes the chain kernel for boundary edge set E , then

$$r_q = \sum_{R \subseteq \binom{[q]}{2}} \text{rank}_{\mathbb{Q}(x)} \left(\sum_{E \supseteq R} (-1)^{|E \setminus R|} M_E(x) \right).$$

For an independent boundary the rank is the number c_q of chains in $C_3^q \setminus \{0^q\}$. We obtain an explicit formula for c_q and

$$c_0 = 1, \quad c_1 = 4, \quad c_2 = 52, \quad c_3 = 1636, \quad \log c_q = 2q \log q + O(q).$$

With arbitrary boundary edges, the first nontrivial full value beyond one terminal is $r_2 = 60$; the finite rank calculation is supplied with an independent certificate and verifier.

At $q = 1$ the general construction collapses to four responses: a rooted piece may or may not supply the root, and may or may not need the root to spend its single force inside the piece. The resulting algebra is

$$\mathbb{Q}(x)[s, a]/(s^2 - s, a^2).$$

This small case is useful computationally. It gives an exact edge transfer, a quadratic tree algorithm, block transfers, a linear clique transfer, and a finite cycle recognizer. For block-cactus graphs the full polynomial is computed in $O(n^2)$ coefficient operations. General tractability of that class also follows from bounded clique-width and MSO machinery; the value of the transfer here is the explicit state system and its concrete arithmetic bound.

The first part of the paper develops the rooted and separator algebras. The second gives the explicit tree, block, clique, and cycle transfers. Finite computations used for r_2 and for the reduced cycle automaton are separated from the symbolic proofs and supplied with independent verifiers.

2. PRELIMINARIES AND RESTRICTED CLOSURE

All graphs are finite and simple. Let $G = (V, E)$ and let $S \subseteq V$. Vertices of S are initially blue and the remaining vertices are white. The *zero-forcing colour-change rule* is

$$u \rightarrow w \quad \text{if } u \text{ is blue and } w \text{ is the unique white neighbour of } u.$$

A set S is a *zero-forcing set* if a sequence of legal forces colours every vertex. We write $z(G; k)$ for the number of zero-forcing sets of cardinality k , and

$$\mathcal{Z}(G; x) = \sum_{k=0}^{|V|} z(G; k) x^k.$$

For a non-empty graph the constant coefficient is zero, but allowing $k = 0$ is convenient for products and for the empty graph.

We shall repeatedly use the elementary observation that a vertex performs at most one force.

Lemma 2.1. *Every vertex is the forcing vertex in at most one step of a zero-forcing process.*

Proof. If u forces w , then immediately after the force every neighbour of u is blue. Since blue vertices never become white, u can never again have a white neighbour. $\square$

Let (H, r) be a rooted graph. We require a closure operation in which the root is prohibited from acting.

Definition 2.2. Let $Q \subseteq V(H)$. For $X \subseteq V(H)$, the Q -restricted closure of X , denoted

$$\text{cl}_H^{-Q}(X),$$

is the terminal blue set obtained by exhausting legal forces while no vertex of Q is permitted to act as a forcing vertex. Vertices of Q may nevertheless be forced. For a singleton $Q = \{r\}$ we write $\text{cl}_H^{-r}(X)$.

The next lemma justifies the notation.

Lemma 2.3 (Confluence and monotonicity). *The set $\text{cl}_H^{-Q}(X)$ is independent of the order in which legal Q -restricted forces are applied. Moreover,*

$$X \subseteq Y \implies \text{cl}_H^{-Q}(X) \subseteq \text{cl}_H^{-Q}(Y).$$

Proof. For $X \subseteq V(H)$ define

$$F_Q(X) = X \cup \{w : \text{there exists } u \in X \setminus Q \text{ with } N_H(u) \setminus X = \{w\}\}.$$

The map F_Q is monotone. Indeed, let $X \subseteq Y$ and suppose that $w \in F_Q(X) \setminus Y$. Then some $u \in X \setminus Q$ has w as its unique neighbour outside X . Since $w \notin Y$, enlarging X to Y does not remove w and cannot introduce a new neighbour outside the blue set. Hence w remains the unique neighbour of u outside Y , so $w \in F_Q(Y)$.

Since $V(H)$ is finite, iteration of F_Q from X reaches the least fixed point of F_Q containing X . Every legal sequential Q -restricted forcing process remains inside this fixed point. Conversely, the vertices introduced at each synchronous iterate may be added in any order by legal Q -restricted forces. The terminal set is therefore unique, and monotonicity follows by iterating the monotone map F_Q . $\square$

For $S \subseteq V(H) \setminus \{r\}$ put

$$C^-(S) = \text{cl}_H^{-r}(S), \quad C^+(S) = \text{cl}_H^{-r}(S \cup \{r\}).$$

The superscript records whether the root is initially white or blue.

Lemma 2.4 (Monotone simulation of a forcing sequence). *Let $B \subseteq B'$ be blue sets in a graph, and let*

$$u_1 \rightarrow w_1, \dots, u_t \rightarrow w_t$$

be a legal forcing sequence from B , with all forcing vertices outside a fixed forbidden set Q . Starting from B' , scan the same list in order and perform $u_i \rightarrow w_i$ whenever w_i is still white, omitting the step otherwise. Every performed step is legal, no forcing vertex lies in Q , and the final blue set contains the final blue set obtained from B .

Proof. Induct on the list. At a step $u \rightarrow w$, if w is already blue there is nothing to prove. Otherwise the current larger blue set contains the current smaller one. The vertex u has no white neighbour in the smaller process other than w ; enlarging the blue set cannot create a new white neighbour. The vertex w remains the unique white neighbour of u , so the step is legal. $\square$

Lemma 2.5 (Source-closure identity). *If $r \in C^-(S)$, then*

$$C^-(S) = C^+(S).$$

Proof. Monotonicity gives $C^-(S) \subseteq C^+(S)$. The set $C^-(S)$ is an r -restricted fixed point and, because it contains r , it contains $S \cup \{r\}$. Since $C^+(S)$ is the least such fixed point, the reverse inclusion follows. $\square$

3. THE FOUR ROOTED RESPONSE STATES

Fix a rooted graph (H, r) and a set $S \subseteq V(H) \setminus \{r\}$.

Definition 3.1 (Source indicator). The source indicator of S is

$$\sigma_r(S) = \begin{cases} 1, & r \in C^-(S), \\ 0, & r \notin C^-(S). \end{cases}$$

Accordingly, $\sigma_r(S) = 1$ precisely when the non-root vertices can force an initially white root without using the root as a forcing vertex.

We next classify the successful behaviour when the root is initially blue.

Definition 3.2 (Passive, active, and blocked states). The set S is called

- (i) *passive* if $C^+(S) = V(H)$;
- (ii) *active* if $C^+(S) \neq V(H)$ and there is a vertex u such that

$$N_H(r) \setminus C^+(S) = \{u\}$$

and

$$\text{cl}_H^{\neg r}(C^+(S) \cup \{u\}) = V(H);$$

- (iii) *blocked* otherwise.

A passive set completes the graph without a force by the root. An active set completes the graph after the root performs its unique available force. A blocked set cannot be completed even if the root is initially blue.

For $\eta \in \{0, 1\}$ define

$$P_\eta(H, r; x) = \sum_{\substack{S \subseteq V(H) \setminus \{r\} \\ S \text{ passive, } \sigma_r(S) = \eta}} x^{|S|},$$

$$A_\eta(H, r; x) = \sum_{\substack{S \subseteq V(H) \setminus \{r\} \\ S \text{ active, } \sigma_r(S) = \eta}} x^{|S|}.$$

We suppress $(H, r; x)$ when no ambiguity can arise, and write

$$P = P_0 + P_1, \quad A = A_0 + A_1.$$

Blocked sets will not enter the transfer algebra because they cannot occur in a successful global configuration.

4. ROOT NORMAL FORM

The following normalization statement makes precise the reordering used throughout the paper.

Lemma 4.1 (Root normalization). *Let (H, r) be rooted, let $S \subseteq V(H) \setminus \{r\}$, and suppose that r is initially blue. Every successful forcing sequence can be replaced by one of the form*

r -restricted saturation ; at most one force by r ; r -restricted saturation.

More precisely, either $C^+(S) = V(H)$, or there is a unique vertex $u \in N_H(r) \setminus C^+(S)$ such that $r \rightarrow u$ is legal from $C^+(S)$ and

$$\text{cl}_H^r(C^+(S) \cup \{u\}) = V(H).$$

Proof. Fix a successful sequence. If the root is never used, the entire sequence is r -restricted, and confluence gives $C^+(S) = V(H)$.

Otherwise let $r \rightarrow u$ be the (necessarily unique) force by r . Replace the prefix preceding this step by any restricted sequence which saturates $S \cup \{r\}$ to $C^+(S)$. The blue set has only increased. If u is still white, every other neighbour of r was blue before the original force and is therefore blue in $C^+(S)$, so $r \rightarrow u$ remains legal. If u has already become blue, omit the force by r . In either case, apply Lemma 2.4 to the remaining suffix. Since r cannot force a second time by Lemma 2.1, every retained force in the suffix is restricted. The resulting sequence has the asserted form and still colours all vertices. $\square$

Theorem 4.2 (Root normal form). *Let (H, r) be a rooted graph and let $S \subseteq V(H) \setminus \{r\}$.*

- (a) *The set $S \cup \{r\}$ is zero forcing in H if and only if S is passive or active.*
- (b) *The set S is zero forcing in H if and only if S is passive or active and $\sigma_r(S) = 1$.*

Consequently,

$$\boxed{\mathcal{Z}(H; x) = (1 + x)(P + A) - (P_0 + A_0).} \quad (1)$$

Proof. Part (a) is exactly Lemma 4.1 together with Definition 3.2. For part (b), a successful process with r initially white must first colour r by a force whose forcing vertex is not r ; hence $r \in C^-(S)$. Conversely, if $\sigma_r(S) = 1$, execute a restricted sequence which colours r . By Lemma 2.5, the resulting restricted closure is $C^+(S)$, after which the passive or active completion from part (a) applies.

Sets containing r contribute $x(P + A)$, whereas sets excluding r contribute $P_1 + A_1 = (P + A) - (P_0 + A_0)$. This proves (1). $\square$

Remark 4.3. The four states record precisely the two pieces of information which can cross a one-vertex boundary: whether the rooted component consumes the root's unique force, and whether it can supply the root when the root is initially white.

5. ROOTED ONE-POINT UNIONS AND DUAL NUMBERS

Let (H, r) and (K, s) be rooted graphs with otherwise disjoint vertex sets. Their *rooted one-point union*, or *wedge*, is the rooted graph

$$(H, r) \vee (K, s)$$

obtained by identifying r and s ; the identified vertex is the root.

Proposition 5.1 (Set-level wedge composition). *Let $S_H \subseteq V(H) \setminus \{r\}$ and $S_K \subseteq V(K) \setminus \{s\}$, and identify $r = s$ to a common root v . Then the following hold.*

- (i) *The wedge is passive exactly when both component sets are passive.*
- (ii) *It is active exactly when one component set is active and the other is passive.*
- (iii) *Every other pair of component states is blocked.*
- (iv) *Its source indicator is the Boolean disjunction of the two component source indicators.*

Proof. With v initially blue and forbidden from forcing, the non-root vertices in H and K have no neighbours across the separator. Thus their restricted closures are computed independently. A passive component completes without using v ; an active component leaves one internal white neighbour which only v can initially force; and a blocked component cannot be completed even when v is supplied and allowed its one force. It follows that a blocked component cannot be rescued by activity on the other side, that two active components cannot both be completed, and that exactly one active component can be completed when every other component is passive. This proves (i)–(iii), using [Lemma 2.1](#).

For (iv), if neither side can force the initially white root, then no first force colouring v exists in the wedge. If one side can source v , execute its restricted source sequence first. By [Lemma 2.5](#), that side then has exactly the same restricted closure as if v had been supplied at time zero. The other side may therefore proceed from its plus closure. In particular, a passive source branch may supply v before an active branch consumes the force of v , and an active source branch may first supply v and then consume that force internally. No additional timing state is needed. $\square$

Theorem 5.2 (Rooted product). *Let*

$$(P^H, P_0^H, A^H, A_0^H) \quad \text{and} \quad (P^K, P_0^K, A^K, A_0^K)$$

be the response data of two rooted graphs. The response data of their rooted wedge satisfy

$$P = P^H P^K, \quad P_0 = P_0^H P_0^K, \quad (2)$$

$$A = A^H P^K + P^H A^K, \quad A_0 = A_0^H P_0^K + P_0^H A_0^K. \quad (3)$$

Proof. Apply [Proposition 5.1](#) to every ordered pair of component subsets and sum their generating weights. The source-zero terms are precisely those in which both component source indicators are zero. $\square$

The product becomes particularly transparent over the algebra of dual numbers. Put

$$\mathbb{D} = \mathbb{Z}[x, \varepsilon]/(\varepsilon^2)$$

and define

$$\begin{aligned} \mathcal{R}(H, r) &= P(H, r) + \varepsilon A(H, r), \\ \mathcal{R}_0(H, r) &= P_0(H, r) + \varepsilon A_0(H, r). \end{aligned}$$

Corollary 5.3 (Dual-number multiplicativity). *For rooted graphs (H, r) and (K, s) ,*

$$\begin{aligned} \mathcal{R}((H, r) \vee (K, s)) &= \mathcal{R}(H, r) \mathcal{R}(K, s), \\ \mathcal{R}_0((H, r) \vee (K, s)) &= \mathcal{R}_0(H, r) \mathcal{R}_0(K, s). \end{aligned}$$

Proof. This is exactly [Theorem 5.2](#), since the coefficient of ε^2 vanishes. $\square$

The four coordinates are not merely sufficient; in the natural connection-matrix sense they are necessary.

Theorem 5.4 (Contextual completeness and rank four). *Work over $K = \mathbb{Q}(x)$ and write*

$$r(H) = (P_0(H), P_1(H), A_0(H), A_1(H))^T$$

for a finite rooted graph H . Let C be the connection matrix indexed by finite rooted graphs, with

$$C_{H,L} = \mathcal{Z}(H \vee L; x).$$

Then $\text{rank}_K C = 4$. Moreover, two rooted graphs have the same response profile if and only if

$$\mathcal{Z}(H \vee L; x) = \mathcal{Z}(H' \vee L; x)$$

for every rooted graph L . It is enough to test the four fixed contexts K_1 , K_2 rooted at an endpoint, and P_3 rooted respectively at an endpoint and at its middle vertex.

Proof. By [Proposition 5.1](#) and [Theorem 4.2](#), compatibility of two successful states is bilinear. In the coordinate order (P_0, P_1, A_0, A_1) ,

$$C_{H,L} = r(H)^T B(x) r(L),$$

where

$$B(x) = \begin{pmatrix} x & x+1 & x & x+1 \\ x+1 & x+1 & x+1 & x+1 \\ x & x+1 & 0 & 0 \\ x+1 & x+1 & 0 & 0 \end{pmatrix}.$$

Indeed, an incompatible pair of active states contributes zero; a compatible pair contributes x when the common root must be initially selected, and an additional 1 exactly when at least one side can source an initially white root. Hence $\text{rank } C \leq 4$.

For the four rooted contexts listed in the statement, direct use of the profiles in [Section 8.1](#) gives the row matrix

$$V(x) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x & 1 & 0 \\ x & x+x^2 & 1 & 0 \\ 0 & x^2 & 0 & 2x \end{pmatrix}.$$

A direct calculation gives

$$\det V = -2x^3, \quad \det B = (x+1)^2.$$

The corresponding 4×4 connection minor is VBV^T and therefore has

$$\det(VBV^T) = 4x^6(x+1)^2 \neq 0.$$

Thus $\text{rank } C = 4$. If the difference of two response vectors pairs to zero with the four displayed contexts, invertibility of B and V forces that difference to vanish. The converse is immediate from the factorization. $\square$

Remark 5.5 (Full response algebra). The complete set-level multiplication can be encoded over K by

$$P_0 \leftrightarrow 1, \quad P_1 \leftrightarrow s, \quad A_0 \leftrightarrow a, \quad A_1 \leftrightarrow sa,$$

with

$$K[s, a]/(s^2 - s, a^2) \cong K[a]/(a^2) \times K[a]/(a^2).$$

The idempotent s records Boolean source disjunction and the nilpotent a records consumption of the root's unique force. Evaluating the source coordinate at $s = 0$ and $s = 1$ recovers respectively the source-zero and total dual-number profiles $\mathcal{R}_0$ and $\mathcal{R}$. Thus the dual-number notation is a pair of projections of the full four-state algebra.

Corollary 5.6 (Scalar specializations). *For the numerical invariant $\mathcal{Z}(G; t)$ over a characteristic-zero field, the rooted connection matrix has rank four when $t \notin \{0, -1\}$, rank two when $t = -1$, and rank zero when $t = 0$.*

Proof. For $t \neq 0$, the displayed matrix $V(t)$ is invertible, so the rank equals that of $B(t)$. This is four except at $t = -1$, where the nonzero part of $B(-1)$ has rank two. At $t = 0$, every rooted wedge is nonempty and its zero-forcing polynomial has zero constant coefficient. $\square$

For a wedge of d rooted graphs the response is therefore

$$P = \prod_{i=1}^d P_i, \quad (4)$$

$$A = \sum_{j=1}^d A_j \prod_{i \neq j} P_i, \quad (5)$$

with the analogous formulas for P_0 and A_0 .

5.1. An exact cut-vertex formula. Let G be connected and let v be a cut vertex. Let $W_1, \dots, W_d$ be the components of $G - v$, and put

$$B_i = G[W_i \cup \{v\}],$$

rooted at v . For each branch let $P_i, P_{i,0}, A_i, A_{i,0}$ denote its response polynomials.

Theorem 5.7 (Articulation formula). *Define*

$$\begin{aligned} R_v(G; x) &= \prod_{i=1}^d P_i + \sum_{j=1}^d A_j \prod_{i \neq j} P_i, \\ R_{v,0}(G; x) &= \prod_{i=1}^d P_{i,0} + \sum_{j=1}^d A_{j,0} \prod_{i \neq j} P_{i,0}. \end{aligned}$$

Then

$$\boxed{\mathcal{Z}(G; x) = (1 + x)R_v(G; x) - R_{v,0}(G; x).} \quad (6)$$

The contribution of zero-forcing sets containing v is $xR_v(G; x)$, and the contribution of those excluding v is $R_v(G; x) - R_{v,0}(G; x)$.

Proof. The rooted graph (G, v) is the wedge of the branches (B_i, v) . Apply [Theorems 4.2](#) and [5.2](#). $\square$

Remark 5.8. Row's cut-vertex method [\[27\]](#) determines the least cardinality of a zero-forcing set by tracking zero spread. Formula [\(6\)](#) is coefficientwise: it reconstructs the number of zero-forcing sets of every cardinality.

6. ARBITRARY SEPARATORS AND ATOMIC PREFORTS

We now pass from a one-vertex boundary to an arbitrary finite separator. The one-terminal response states remain useful later as selection projectors, but the natural general interface is most transparent in the fort language. Throughout this section $B = [q] = \{1, \dots, q\}$ is an ordered boundary and the corresponding vertices of a boundaried graph are denoted $b_1, \dots, b_q$.

Definition 6.1 (Boundaried graph and gluing). A q -boundaried graph is a finite simple graph H together with pairwise distinct labeled vertices $b_1, \dots, b_q$. If H and K are q -boundaried graphs, $H \oplus_q K$ is obtained from their disjoint union by identifying equally labeled boundary vertices and

suppressing duplicate edges. Let $E_B(H) \subseteq \binom{B}{2}$ denote the graph induced by the boundary.

The q th connection matrix of the zero-forcing polynomial is the infinite matrix

$$C_q[H, K] = \mathcal{Z}(H \oplus_q K; x)$$

over all isomorphism classes of q -boundaried graphs. We write

$$r_q = \text{rank}_{\mathbb{Q}(x)} C_q.$$

Unless a specialization is stated explicitly, every connection rank in this paper is taken over the rational-function field $\mathbb{Q}(x)$. For $q = 0$, multiplicativity on disjoint unions gives $r_0 = 1$. Theorem 5.4 gives $r_1 = 4$.

6.1. Preforts and atomic words. Recall that a fort is a nonempty set F such that no vertex outside F has exactly one neighbour in F . Equivalently, a set is zero forcing if and only if it meets every fort [1]. We omit the boundary conditions locally.

Definition 6.2 (Prefort). Let H be q -boundaried and let $S \subseteq V(H) \setminus B$ be an interior initial set. A nonempty set $F \subseteq V(H) \setminus S$ is a B -prefort if

$$|N_H(v) \cap F| \neq 1 \quad \text{for every } v \in V(H) \setminus (F \cup B).$$

The fort condition is imposed at every nonboundary vertex and omitted at B .

Write l for the symbol recording boundary membership. For a prefort F and $i \in B$ put

$$c_i(F) = |N_H(b_i) \cap (F \setminus B)|.$$

Its set of *atomic refinements* consists of all words $\alpha \in \{0, 1, \mathsf{l}\}^q$ whose i th coordinate is

$$\alpha_i = \begin{cases} \mathsf{l}, & b_i \in F, \\ 0, & b_i \notin F \text{ and } c_i(F) = 0, \\ 1, & b_i \notin F \text{ and } c_i(F) = 1, \\ 0 \text{ or } 1, & b_i \notin F \text{ and } c_i(F) \geq 2. \end{cases}$$

The last line is the crucial saturation rule: once two interior fort neighbours are already present, adding an external contribution can never make the final count equal to one, so either binary refinement is safe for an existential compatibility test.

Let

$$\mathcal{A}_H(S)$$

be the union of the atomic refinements of all B -preforts disjoint from S . The all-zero atom has a special status.

Lemma 6.3 (Fatal atom). *If $0^q \in \mathcal{A}_H(S)$, then S cannot occur as the interior part of a zero-forcing set in $H \oplus_q K$ for any external q -boundaried graph K and any choice of initially blue boundary vertices.*

Proof. The atom 0^q is supplied by a nonempty prefort F containing no boundary vertex and for which every boundary vertex has either zero or at least two neighbours in F . Hence the omitted fort conditions at the boundary are already satisfied. It follows that F is a fort of H , and it remains a fort after any external graph is glued along B , because no new vertex of F is introduced and external vertices have no neighbours in F except through boundary vertices, none of which lies in F . Since $F \cap S = \emptyset$, no extension of S is zero forcing. $\square$

We therefore work with the nonfatal atom poset

$$P_q = \{0, 1, \mathsf{l}\}^q \setminus \{0^q\},$$

ordered coordinatewise by

$$0 < 1 < \mathsf{l}.$$

The join $\alpha \vee \beta$ is the coordinatewise maximum.

Lemma 6.4 (Join closure). *If S is nonfatal, then $\mathcal{A}_H(S) \subseteq P_q$ is join closed:*

$$\alpha, \beta \in \mathcal{A}_H(S) \implies \alpha \vee \beta \in \mathcal{A}_H(S).$$

Proof. Choose prefors F, G and atomic refinements producing α, β . The union $F \cup G$ is again a prefort. Indeed, if a nonboundary vertex lies outside $F \cup G$, then its number of neighbours in each of F and G is zero or at least two; the same is therefore true for the union.

At a boundary coordinate, membership in either prefort gives the symbol l in the union. If neither contains the boundary vertex and at least one chosen atom has value 1, then the union has at least one interior prefort neighbour; if the actual count is one the refinement is forced to be 1, and if it is at least two the refinement 1 is available. If both chosen values are zero, the union either has count zero or at least two, in both cases allowing the refinement zero. The union $F \cup G$ supplies $\alpha \vee \beta$. $\square$

Let $\mathfrak{J}_q$ denote the finite set of join-closed subsets of P_q , including the empty set. Every nonfatal decorated piece (H, S) therefore has a state

$$\Theta(H, S) = (E_B(H), \mathcal{A}_H(S)) \in 2^{\binom{B}{2}} \times \mathfrak{J}_q.$$

6.2. The exact gluing criterion. For an atom α define its trace

$$T(\alpha) = \{i \in B : \alpha_i = \mathsf{l}\}.$$

Let $E \subseteq \binom{B}{2}$ be a boundary-edge graph and put

$$d_E(i, T) = |N_E(i) \cap T|.$$

Definition 6.5 (Atomic compatibility). Atoms $\alpha, \beta \in P_q$ are *E-compatible* if they have the same trace T and

$$d_E(i, T) + \alpha_i + \beta_i \neq 1 \quad (i \notin T), \quad (7)$$

where outside T the symbols are the integers 0, 1.

Theorem 6.6 (Prefort gluing). *Let H, K be q -boundaried graphs with interior initial sets S_H, S_K , and let*

$$E = E_B(H) \cup E_B(K).$$

There is a fort of $H \oplus_q K$ disjoint from $S_H \cup S_K$ with boundary trace T if and only if there exist E -compatible atoms

$$\alpha \in \mathcal{A}_H(S_H), \quad \beta \in \mathcal{A}_K(S_K)$$

with common trace T .

Proof. Suppose first that F is a global fort. Its restrictions $F_H = F \cap V(H)$ and $F_K = F \cap V(K)$ are local preforts and agree on the boundary trace. At a boundary vertex outside the trace, the total number of fort neighbours is

$$c_i(F_H) + c_i(F_K) + d_E(i, T),$$

and this number is not one. If a local count is zero or one, take the forced atomic value. If it is at least two, choose a binary refinement so that (7) holds; this is always possible because the actual count is already at least two. Compatible refinements exist.

Conversely, choose local preforts and atomic refinements producing compatible α, β . Their union has the correct fort condition at every nonboundary vertex by the definition of prefort. The traces agree, so the union is well defined at B . At $i \notin T$, if both local counts are zero or one, equation (7) is exactly the global fort condition. If either local count is at least two, the global count is at least two automatically. Hence the union is a global fort. $\square$

For $A, D \in \mathfrak{J}_q$ let

$$\mathcal{T}_E(A, D) = \{T : \text{an } E\text{-compatible pair in } A \times D \text{ has trace } T\}.$$

If $X \subseteq B$ is the initially blue part of the boundary, a global fort disjoint from the initial set can only have trace disjoint from X . We obtain the separator kernel

$$K_E(A, D; x) = \sum_{\substack{X \subseteq B \\ X \cap T \neq \emptyset \ \forall T \in \mathcal{T}_E(A, D)}} x^{|X|}. \quad (8)$$

If $\emptyset \in \mathcal{T}_E(A, D)$ the sum is zero; if $\mathcal{T}_E(A, D) = \emptyset$, it is $(1 + x)^q$.

Example 6.7 (A complete two-terminal calculation). Let $q = 2$, write the independent boundary as $B = \{a, b\}$, and consider the two decorated pieces in Figure 1. In the left piece H , the interior vertices are v, c , the edges are

$$av, bv, ac, bc,$$

and $S_H = \{c\}$. Since the selected vertex c lies outside every prefort disjoint from S_H , its two neighbours a, b must occur together or not at all. The nonempty preforts are therefore

$$\{v\}, \quad \{a, b\}, \quad \{a, b, v\}.$$

They supply, respectively, the atoms $11, \mathbb{I}$, and $\mathbb{I}$. Thus

$$\mathcal{A}_H(S_H) = \{11, \mathbb{I}\},$$

a two-element chain. In the right piece K , the only interior vertex w is selected and adjacent to both boundary vertices. Its prefort condition excludes $\{a\}$ and $\{b\}$, so

$$\mathcal{A}_K(S_K) = \{\mathbb{I}\}, \quad S_K = \{w\}.$$

For an independent boundary, two atoms are compatible exactly when they are equal. Hence the only compatible pair is $\mathbb{I}/\mathbb{I}$, with trace $\{a, b\}$. A selected boundary set must meet this trace, and therefore

$$K_{\emptyset}(\mathcal{A}_H(S_H), \mathcal{A}_K(S_K); x) = 2x + x^2.$$

Equivalently, the chain formula below gives the same value directly: $\mathcal{A}_H(S_H) \cap \mathcal{A}_K(S_K) = \{\mathbb{I}\}$, so the empty-chain term contributes $(1+x)^2$ and the singleton-chain term contributes -1 . Thus $(1+x)^2 - 1 = 2x + x^2$. The fixed interior choice $S_H \cup S_K = \{c, w\}$ consequently contributes

$$x^2(2x + x^2) = 2x^3 + x^4$$

to the polynomial of the glued graph. Directly, one must initially select at least one of a, b ; either choice starts the forcing process, while selecting neither leaves both boundary vertices white and no force is available.

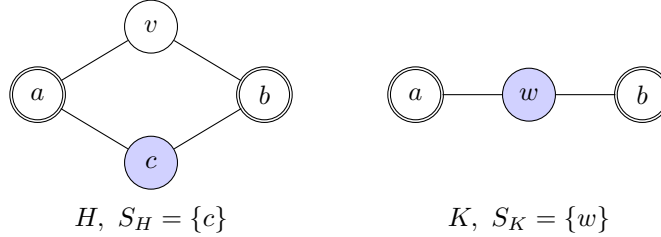


FIGURE 1. Worked $q = 2$ separator example. Double circles are boundary vertices; shaded vertices are prescribed members of the interior initial set. The left atom family is the chain $\{11, \mathbb{I}\}$ and the right family is $\{\mathbb{I}\}$.

Theorem 6.8 (Separator splitting formula). *Let H, K be q -boundaried graphs, write $E = E_B(H) \cup E_B(K)$, and for $A \in \mathfrak{J}_q$ define*

$$R_H(A; x) = \sum_{\substack{S \subseteq V(H) \setminus B \\ 0^q \notin \mathcal{A}_H(S), \mathcal{A}_H(S) = A}} x^{|S|},$$

with R_K defined analogously. Then

$$\boxed{\mathcal{Z}(H \oplus_q K; x) = \sum_{A, D \in \mathfrak{J}_q} R_H(A; x) K_E(A, D; x) R_K(D; x).} \quad (9)$$

Proof. Partition an initial set of $H \oplus_q K$ uniquely into its interior parts S_H, S_K and its selected boundary set X . Fatal interior states cannot succeed by [Lemma 6.3](#). For a nonfatal pair, the fort characterization and [Theorem 6.6](#) say that the total initial set is zero forcing exactly when X intersects every trace in $\mathcal{T}_E(\mathcal{A}_H(S_H), \mathcal{A}_K(S_K))$. Summing the weights $x^{|S_H|+|S_K|+|X|}$ gives (9). $\square$

We obtain a splitting formula for every finite separating set. The rest of this section determines the linear information actually needed by this finite but potentially large kernel.

7. CHAIN COMPRESSION AND EXACT CONNECTION RANKS

Let $\mathcal{C}_q$ be the set of chains of P_q , including the empty chain, and put

$$c_q = |\mathcal{C}_q|.$$

The key cancellation is independent of zero forcing.

Lemma 7.1 (Chain disjointness identity). *If $A, D \in \mathfrak{J}_q$, then*

$$\mathbf{1}[A \cap D = \emptyset] = \sum_{\substack{C \in \mathcal{C}_q \\ C \subseteq A \cap D}} (-1)^{|C|}. \quad (10)$$

Proof. Ordinary inclusion–exclusion gives

$$\mathbf{1}[A \cap D = \emptyset] = \sum_{X \subseteq A \cap D} (-1)^{|X|}.$$

For a finite nonempty join-semilattice J , call $j \in J$ *join-irreducible* if there do not exist $u, v < j$ with $j = u \vee v$. Let $I(J)$ be the set of join-irreducible elements. Two elementary finite semilattice facts will be used explicitly.

First, $I(J)$ generates J . Indeed, proceed by induction upward from the minimal elements. A minimal element is join-irreducible. If j is not join-irreducible, write $j = u \vee v$ with $u, v < j$ and apply the induction hypothesis to u and v . Second, every generating set X for J must contain $I(J)$. If $j \in I(J)$ is a join of finitely many elements of X , repeated use of join-irreducibility forces j to equal one of those generators.

Now group the subsets $X \subseteq A \cap D$ according to the join-closed set $J = \langle X \rangle_\vee$ that they generate. For nonempty J , the preceding facts show that the generating subsets are exactly

$$I(J) \cup Y, \quad Y \subseteq J \setminus I(J).$$

Their total inclusion–exclusion coefficient is therefore

$$\sum_{Y \subseteq J \setminus I(J)} (-1)^{|I(J)|+|Y|} = (-1)^{|I(J)|} (1-1)^{|J|-|I(J)|}.$$

This is nonzero precisely when every element of J is join-irreducible. That happens precisely when J is a chain. If $a, b \in J$ are incomparable, then $a \vee b$ is strictly larger than both and is therefore reducible. Conversely, in a chain

the join of two elements is their maximum, so no element can be the join of two strictly smaller elements. A nonempty chain C contributes $(-1)^{|C|}$.

Finally, $X = \emptyset$ generates the empty join-closed set; its coefficient is 1, which is exactly the contribution of the empty chain in (10). This proves the identity including the empty case. $\square$

Define the chain-incidence matrix

$$Z[A, C] = \mathbf{1}[C \subseteq A], \quad A \in \mathfrak{J}_q, \quad C \in \mathcal{C}_q.$$

When its rows are restricted to chain states, Z is the zeta matrix of the finite poset $\mathcal{C}_q$ ordered by inclusion. After a linear extension it is triangular with diagonal one, hence invertible.

The next lemma is what allows chain compression to survive arbitrary boundary edges.

Lemma 7.2 (Compatibility neighbourhoods are join closed). *Fix a boundary-edge graph E , a selected boundary set X , and $D \in \mathfrak{J}_q$. Let $N_{E,X}(D)$ be the set of atoms $\alpha \in P_q$ whose trace is disjoint from X and for which some $\beta \in D$ is E -compatible with α . Then $N_{E,X}(D)$ is join closed.*

Proof. Take $\alpha_1, \alpha_2 \in N_{E,X}(D)$ with compatible witnesses $\beta_1, \beta_2 \in D$. Since D is join closed, $\beta = \beta_1 \vee \beta_2$ belongs to D ; put $\alpha = \alpha_1 \vee \alpha_2$. Their common trace is the union of the two original traces and is still disjoint from X .

Suppose compatibility failed at i outside that union. If the boundary-edge contribution $d_E(i, T)$ were one and both joined bits zero, then one of the original traces already contributed that unique boundary neighbour while the corresponding original bits were zero, contradicting compatibility of that pair. If $d_E(i, T) = 0$ and the two joined bits summed to one, then one joined bit, say on the α side, is one while the other is zero. Some original α_j therefore has bit one, every β_j has bit zero, and no original trace has an E -neighbour at i ; again the j th pair would violate (7). Contributions at least two are automatically safe. The atoms α and β are compatible. $\square$

For fixed E and X , the success indicator is therefore

$$k_{E,X}(A, D) = \mathbf{1}[A \cap N_{E,X}(D) = \emptyset].$$

By Lemma 7.1, its dependence on A lies in the column space of Z . By symmetry the same is true in the D coordinate. Summing $x^{|X|} k_{E,X}$ over X proves the following.

Theorem 7.3 (Chain compression). *For every boundary-edge graph E , the matrix*

$$(K_E(A, D; x))_{A, D \in \mathfrak{J}_q}$$

over $\mathbb{Q}(x)$ has the same rank as its submatrix obtained by restricting both rows and columns to chain states $C, D \in \mathcal{C}_q$. In particular its rank is at most c_q .

Proof. The preceding discussion gives a factorization $K_E = ZH_E$ for some matrix H_E . Since the restriction of Z to chain rows is invertible, the chain rows span the full row space. Applying the same argument to the symmetric matrix K_E shows that the chain columns span the full column space. Hence the chain-chain submatrix has the same rank. $\square$

7.1. Independent boundaries and chain enumeration. When $E = \emptyset$, two atoms are compatible exactly when they are equal. Indeed they must have the same trace, and outside the trace $\alpha_i + \beta_i \neq 1$ is equivalent to $\alpha_i = \beta_i$ for binary bits. Inclusion-exclusion over the common atoms gives the more explicit factorization

$$K_{\emptyset}(A, D; x) = \sum_{\substack{C \in \mathcal{C}_q \\ C \subseteq A \cap D}} (-1)^{|C|} (1+x)^{q-|T(C)|}, \quad (11)$$

where $T(C) = \bigcup_{\alpha \in C} T(\alpha)$. Thus

$$K_{\emptyset} = ZW(x)Z^T, \quad W_{C,C}(x) = (-1)^{|C|} (1+x)^{q-|T(C)|}.$$

Every diagonal entry is nonzero in $\mathbb{Q}(x)$ and the chain rows of Z are invertible. Therefore

$$\text{rank}_{\mathbb{Q}(x)} K_{\emptyset} = c_q. \quad (12)$$

The numbers c_q are explicit.

Proposition 7.4 (Chain count). *For $q \geq 0$,*

$$c_q = \frac{1}{2} \left[1 + \sum_{k=1}^{2q+1} \sum_{j=1}^k (-1)^{k-j} \binom{k-1}{j-1} \binom{j+2}{2}^q \right]. \quad (13)$$

Moreover

$$\log c_q = 2q \log q + O(q).$$

Proof. A weak j -term chain in C_3^q is chosen independently in every coordinate. The number of nondecreasing j -tuples from a three-element chain is $\binom{j+2}{2}$, so there are $\binom{j+2}{2}^q$ weak chains. If $s_q(k)$ is the number of strict k -element chains, then a weak j -chain with exactly k distinct terms is obtained by choosing the $k-1$ change positions, giving

$$\binom{j+2}{2}^q = \sum_{k=1}^j \binom{j-1}{k-1} s_q(k).$$

Binomial inversion yields

$$s_q(k) = \sum_{j=1}^k (-1)^{k-j} \binom{k-1}{j-1} \binom{j+2}{2}^q.$$

A strict chain in C_3^q has at most $2q+1$ elements. Since C_3^q has a unique minimum 0^q , adding or deleting that minimum bijects chains avoiding it

with chains containing it. Hence the number of chains in $P_q = C_3^q \setminus \{0^q\}$ is one half of the total number of chains in C_3^q , proving (13).

For the asymptotics, maximal chains are obtained by ordering the $2q$ labeled coordinate increments, subject only to the first increment of each coordinate preceding its second. There are $(2q)!/2^q$ such chains. Conversely, encode an arbitrary chain by assigning each of the $2q$ coordinate increments a time in $\{0, 1, \dots, 2q+1\}$, where the extreme values mean that the increment occurs before the first or after the last displayed element. This gives the upper bound $(2q+2)^{2q}$. Stirling's formula now gives the stated logarithmic order. $\square$

The first values are

q	0	1	2	3	4	5	6	7
c_q	1	4	52	1636	95668	8977444	1234728052	234059341156.

7.2. Realizing every chain state. The chain compression above is an upper-bound statement unless its chain modes can be produced by actual graph pieces. We next give an explicit realization. A *decorated graph* means a graph together with a prescribed interior initial set S ; selected constraint vertices below belong to S and therefore cannot lie in a prefort disjoint from S .

Lemma 7.5 (Equality and implication gadgets). *Let u, v be unselected nonboundary vertices.*

- (a) *A selected vertex adjacent exactly to u, v enforces $u \in F \iff v \in F$ in every prefort F disjoint from the selected set.*
- (b) *After adding an unselected duplicate v' equality-tied to v , a selected vertex adjacent to u, v, v' enforces $u \in F \implies v \in F$.*

Proof. A selected constraint vertex lies outside F , so its prefort condition says that the number of its neighbours in F is not one. For degree two this allows only 00 and 11, proving equality. In part (b), the duplicate has the same membership as v , so the three-neighbour counts for (u, v) are 0, 2, 1, 3 on 00, 01, 10, 11, respectively. Exactly the assignment 10 is forbidden. $\square$

Theorem 7.6 (Chain realization). *For every chain $C \in \mathcal{C}_q$ and every desired boundary-edge set $E \subseteq \binom{B}{2}$, there exists a finite simple q -boundaried graph H_C and an interior initial set S_C such that*

$$E_B(H_C) = E, \quad 0^q \notin \mathcal{A}_{H_C}(S_C), \quad \mathcal{A}_{H_C}(S_C) = C.$$

Proof. We give the construction with a formal vertex and edge description. All vertices called *selected* below belong to S_C ; all other auxiliary vertices are unselected.

The empty chain. For $C = \emptyset$, take the boundary B and, for every i , add a selected leaf p_i adjacent only to b_i . If a prefort disjoint from S_C contained b_i , then the selected vertex p_i , which lies outside that prefort, would have exactly one prefort neighbour. Hence every boundary vertex is excluded.

There are no other unselected vertices, so no nonempty prefort exists and the atomic family is empty. The prescribed boundary edges E may then be added.

The nonempty chain: master variables. Write

$$C = \{\alpha^1 < \alpha^2 < \dots < \alpha^k\}, \quad k \geq 1.$$

Introduce unselected master vertices $t_1, \dots, t_k$. For every $1 \leq j < k$, introduce an unselected duplicate d_j and selected vertices e_j, p_j , with edges

$$e_j t_j, e_j d_j, \quad p_j t_{j+1}, p_j t_j, p_j d_j.$$

The degree-two selected vertex e_j enforces $d_j \in F \iff t_j \in F$, and then the degree-three selected vertex p_j enforces

$$t_{j+1} \in F \implies t_j \in F$$

by [Lemma 7.5](#). Hence the possible membership vectors of the master vertices are exactly the prefixes

$$00 \dots 0, 10 \dots 0, 110 \dots 0, \dots, 11 \dots 1.$$

Encoding one boundary coordinate. Fix $i \in B$. Because C is a chain, the coordinate sequence $\alpha_i^1, \dots, \alpha_i^k$ is nondecreasing in $0 < 1 < l$. If the value 1 occurs, let

$$a(i) = \min\{j : \alpha_i^j = 1\}.$$

Introduce an unselected marker m_i , join it to b_i , and introduce a selected equality vertex f_i with edges $f_i m_i, f_i t_{a(i)}$. Thus $m_i \in F$ exactly from prefix $a(i)$ onward. If the value 1 never occurs, no marker is introduced.

If the value l occurs, let

$$\iota(i) = \min\{j : \alpha_i^j = l\}.$$

Introduce a selected equality vertex g_i with edges $g_i b_i, g_i t_{\iota(i)}$; hence b_i enters the prefort exactly from prefix $\iota(i)$ onward. If l never occurs, instead introduce a selected pin h_i adjacent only to b_i , which forces b_i to remain outside every prefort. A direct jump $0 \rightarrow l$ requires no marker.

Let S_C be the set of all selected vertices

$$\{e_j, p_j\}_{j < k} \cup \{f_i, g_i, h_i : \text{created above}\},$$

and let H_C contain precisely the boundary, master, duplicate, marker, and selected constraint vertices just described, together with the listed edges. Finally add the prescribed edge set E inside the boundary. These boundary edges do not enter any nonboundary prefort condition.

No unintended preforts. Let F be a prefort disjoint from S_C . The equality and implication constraints force the master membership set to be a prefix $\{t_1, \dots, t_s\}$ for some $0 \leq s \leq k$, and all duplicates, markers, and unpinned boundary vertices are then determined by equality. If $s = 0$, every duplicate and marker is outside F , every equality-tied boundary vertex is outside F , and every remaining boundary vertex is pinned out; hence F is empty, contrary to the definition of prefort. Thus $1 \leq s \leq k$.

Conversely, each nonzero prefix gives a prefort. Every selected equality or implication vertex sees one of the allowed neighbour counts 0, 2, or 3 by construction. An unselected master vertex or duplicate that lies outside the prefort is adjacent only to selected constraint vertices and therefore sees zero prefort neighbours. The only additional check concerns a marker m_i outside the prefort. This means $s < a(i)$. Monotonicity of the coordinate sequence implies that b_i has not yet reached 1, so b_i is also outside the prefort. The marker therefore sees zero prefort neighbours (its other neighbour is the selected equality vertex f_i). Boundary vertices need not satisfy a local fort condition by definition of prefort. Hence no unselected auxiliary vertex creates an unintended restriction.

For prefix s , coordinate i is 0 before $a(i)$, is 1 from $a(i)$ until $\iota(i) - 1$ when these stages exist, and is 1 from $\iota(i)$ onward. The resulting atomic word is exactly α^s . Each boundary vertex has at most one interior prefort neighbour, so no ≥ 2 count creates extra wildcard refinements. Therefore

$$\mathcal{A}_{H_C}(S_C) = \{\alpha^1, \dots, \alpha^k\} = C.$$

Since $C \subseteq P_q$, the fatal atom 0^q does not occur. Adding the edges of E inside B changes neither the prefort conditions away from B nor the interior counts $c_i(F)$, completing the construction. $\square$

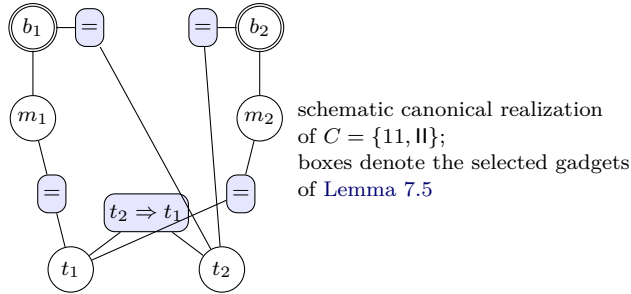


FIGURE 2. Schematic of the chain-realization construction for a representative $q = 2$ chain. The implication box abbreviates the duplicate-plus-constraint gadget specified formally in the proof; equality boxes are selected degree-two constraint vertices.

7.3. Selection projectors. To turn a decorated realization into a row of the ordinary connection matrix, we must isolate the prescribed initial set. The rank-four one-terminal calculus provides explicit projectors. We work with formal $\mathbb{Q}(x)$ -linear combinations of rooted graphs, usually called quantum graphs in the connection-matrix literature.

Let $K_1, K_2, P_3^{\text{end}}, P_3^{\text{mid}}$ denote the four rooted contexts used in [Theorem 5.4](#). Define

$$Q_{\text{in}} = -\frac{1}{x+1}K_1 - \frac{1}{x(x+1)}K_2 + \frac{1}{x(x+1)}P_3^{\text{end}}, \quad (14)$$

$$Q_{\text{out}} = \frac{x+2}{x+1}K_1 + \frac{1}{x(x+1)}K_2 - \frac{1}{x(x+1)}P_3^{\text{end}}. \quad (15)$$

Lemma 7.7 (Vertex-selection projectors). *For every rooted graph (G, v) ,*

$$\begin{aligned} \mathcal{Z}(G \vee Q_{\text{in}}; x) &= \sum_{\substack{S \text{ zero forcing in } G \\ v \in S}} x^{|S|}, \\ \mathcal{Z}(G \vee Q_{\text{out}}; x) &= \sum_{\substack{S \text{ zero forcing in } G \\ v \notin S}} x^{|S|}. \end{aligned}$$

The identities remain valid when G contains arbitrary additional labeled boundary vertices and arbitrary external attachments away from the interior of the projector.

Proof. Write $r(G) = (P_0, P_1, A_0, A_1)^T$. The rooted profiles used in the projectors are

$$r(K_1) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad r(K_2) = \begin{pmatrix} 0 \\ x \\ 1 \\ 0 \end{pmatrix}, \quad r(P_3^{\text{end}}) = \begin{pmatrix} x \\ x+x^2 \\ 1 \\ 0 \end{pmatrix}.$$

Consequently the formal response vectors of the two quantum graphs simplify to

$$r(Q_{\text{in}}) = \begin{pmatrix} 0 \\ x/(x+1) \\ 0 \\ 0 \end{pmatrix}, \quad r(Q_{\text{out}}) = \begin{pmatrix} 1 \\ -x/(x+1) \\ 0 \\ 0 \end{pmatrix}.$$

Multiplication by the connection matrix $B(x)$ of [Theorem 5.4](#) gives

$$B(x)r(Q_{\text{in}}) = \begin{pmatrix} x \\ x \\ x \\ x \end{pmatrix}, \quad B(x)r(Q_{\text{out}}) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Hence

$$r(G)^T B(x)r(Q_{\text{in}}) = x(P_0 + P_1 + A_0 + A_1), \quad r(G)^T B(x)r(Q_{\text{out}}) = P_1 + A_1,$$

which are exactly the contributions from initial sets containing and omitting v , respectively. All calculations take place over $\mathbb{Q}(x)$, so the denominators in the projectors are legitimate; specializations such as $x = 0, -1$ are separate questions.

The argument is coefficientwise in any additional vertex-set variables on $G - v$. Thus extra labeled boundary vertices or external attachments can be

held fixed as part of the rooted context, and successive projectors at distinct vertices act as commuting membership filters. $\square$

Applying these projectors successively at distinct vertices filters the multivariate zero-forcing generating function coefficientwise, so they commute as membership filters.

Corollary 7.8 (Pinning a prescribed interior set). *Let H be a q -boundaried graph, $U = V(H) \setminus B$, and $S \subseteq U$. There is a q -boundaried quantum graph $\Pi_{H,S}$ which, in every external q -boundaried context, retains exactly the zero-forcing sets whose intersection with U is S . In particular, if (H, S) realizes a chain state C and (K, T) realizes a chain state D , then*

$$\mathcal{Z}(\Pi_{H,S} \oplus_q \Pi_{K,T}; x) = x^{|S|+|T|} K_{E_B(H) \cup E_B(K)}(C, D; x).$$

Proof. Attach Q_{in} at every vertex of S and Q_{out} at every vertex of $U \setminus S$. Repeated application of Lemma 7.7 retains exactly the desired membership pattern, and the retained interior vertices contribute $x^{|S|}$. Applying the same construction independently to (K, T) and then using the separator splitting theorem leaves only the prescribed pair of local states, giving the displayed pairing identity. $\square$

7.4. The master connection-rank formula. For a fixed boundary-edge graph E , let

$$M_E(x) = (K_E(C, D; x))_{C, D \in \mathcal{C}_q} \in \mathbb{Q}(x)^{c_q \times c_q}$$

be the chain kernel. Put $m = \binom{q}{2}$. The full finite *chain-edge connection kernel* $\mathbb{M}_q \in \mathbb{Q}(x)^{2^m c_q \times 2^m c_q}$ is indexed by pairs (F, C) with $F \subseteq \binom{B}{2}$ and $C \in \mathcal{C}_q$ and has block

$$\mathbb{M}_q[(F, C), (G, D)] = M_{F \cup G}(x)[C, D].$$

Theorem 7.9 (Exact finite connection model). *For every $q \geq 0$,*

$$\boxed{r_q = \text{rank}_{\mathbb{Q}(x)} \mathbb{M}_q.}$$

The infinite connection matrix of the zero-forcing polynomial has exactly the same rank as the explicit finite chain-edge kernel.

Proof. Put $m = \binom{q}{2}$ and $j_q = |\mathfrak{J}_q|$. We prove the two inequalities separately and keep the finite and infinite column spaces distinct throughout.

Upper bound. For a fixed glued boundary-edge graph E , let

$$U_E = (K_E(A, D; x))_{A, D \in \mathfrak{J}_q} \in \mathbb{Q}(x)^{j_q \times j_q}.$$

Let $Z \in \{0, 1\}^{j_q \times c_q}$ be the join-state/chain incidence matrix $Z[A, C] = \mathbf{1}[C \subseteq A]$, and let Z_0 be its square submatrix on chain rows. As observed above, Z_0 is the zeta matrix of chains ordered by inclusion and is invertible; in particular Z has full column rank.

The proof of [Theorem 7.3](#) shows that the column and row spaces of U_E are contained in the column and row spaces generated by Z . Choose any left inverse L of Z and set

$$N_E = LU_EL^T \in \mathbb{Q}(x)^{c_q \times c_q}.$$

Since ZL acts as the identity on the column space of U_E and $L^T Z^T$ acts as the identity on its row space,

$$U_E = ZN_E Z^T. \quad (16)$$

Restricting (16) to chain rows and columns gives

$$M_E = Z_0 N_E Z_0^T. \quad (17)$$

The matrices N_E and M_E are congruent through the invertible matrix Z_0 .

Let $\mathbb{U}_q$ be the universal finite matrix indexed by local states (F, A) with $F \subseteq \binom{B}{2}$ and $A \in \mathfrak{J}_q$, whose (F, G) block is $U_{F \cup G}$. Let $\mathbb{N}_q$ be the analogous block matrix with blocks $N_{F \cup G}$. Equations (16) and (17) give the explicit factorizations

$$\mathbb{U}_q = (I_{2^m} \otimes Z) \mathbb{N}_q (I_{2^m} \otimes Z)^T, \quad (18)$$

$$\mathbb{M}_q = (I_{2^m} \otimes Z_0) \mathbb{N}_q (I_{2^m} \otimes Z_0)^T. \quad (19)$$

Because Z has full column rank and Z_0 is invertible,

$$\text{rank } \mathbb{U}_q = \text{rank } \mathbb{N}_q = \text{rank } \mathbb{M}_q.$$

For each ordinary q -boundaried graph H , form its finite response vector ρ_H indexed by (F, A) : it is zero unless $F = E_B(H)$, and at $(E_B(H), A)$ it equals the polynomial $R_H(A; x)$ from [Theorem 6.8](#). The splitting formula is precisely

$$C_q[H, K] = \rho_H^T \mathbb{U}_q \rho_K.$$

Hence the infinite connection matrix factors through $\mathbb{U}_q$, and

$$r_q \leq \text{rank } \mathbb{U}_q = \text{rank } \mathbb{M}_q.$$

Lower bound. Index the chain-edge states by $i = (F, C)$. By [Theorem 7.6](#), choose for each i a decorated realization (H_i, S_i) with boundary-edge set F and chain state C , and put

$$Q_i = \Pi_{H_i, S_i}.$$

Apply the pinning construction on *both* sides. If $i = (F, C)$ and $j = (G, D)$, [Corollary 7.8](#) gives

$$\mathcal{Z}(Q_i \oplus_q Q_j; x) = x^{|S_i| + |S_j|} K_{F \cup G}(C, D; x). \quad (20)$$

Therefore the finite pairing matrix of the quantum graphs Q_i is

$$(\mathcal{Z}(Q_i \oplus_q Q_j; x))_{i,j} = D \mathbb{M}_q D, \quad D_{ii} = x^{|S_i|}. \quad (21)$$

The diagonal matrix D is invertible over $\mathbb{Q}(x)$.

Each Q_i is a finite $\mathbb{Q}(x)$ -linear combination of ordinary q -boundaried graphs. Taking the finite union of all ordinary graphs that occur in these

combinations, let P be their coefficient matrix and let C_q^{fin} be the corresponding finite submatrix of the ordinary connection matrix. Bilinearity of gluing gives

$$D\mathbb{M}_q D = P C_q^{\text{fin}} P^T.$$

Consequently

$$\text{rank } \mathbb{M}_q = \text{rank}(D\mathbb{M}_q D) \leq \text{rank } C_q^{\text{fin}} \leq r_q.$$

Together with the upper bound, this proves the theorem. $\square$

The Boolean transform makes this rank additive over edge sectors. For $R \subseteq \binom{B}{2}$ define

$$\widehat{M}_R(x) = \sum_{E \supseteq R} (-1)^{|E \setminus R|} M_E(x). \quad (22)$$

Theorem 7.10 (Master rank formula). *For every $q \geq 0$,*

$$r_q = \sum_{R \subseteq \binom{[q]}{2}} \text{rank}_{\mathbb{Q}(x)} \widehat{M}_R(x). \quad (23)$$

In particular, if r_q° denotes the connection rank of the submatrix obtained by restricting both local sides to q -boundaried graphs whose boundary induces no edge, then

$$r_q^\circ = c_q. \quad (24)$$

Consequently

$$c_q \leq r_q \leq 2^{\binom{q}{2}} c_q. \quad (25)$$

Proof. Let $L = \binom{B}{2}$ and index the subsets of L in any linear extension of inclusion. Define the Boolean zeta matrix

$$Z_{\text{edge}}[F, R] = \mathbf{1}[F \subseteq R], \quad F, R \subseteq L.$$

It is triangular with diagonal one and therefore invertible. By the definition (22), Boolean Möbius inversion gives

$$M_E = \sum_{R \supseteq E} \widehat{M}_R.$$

Hence the (F, G) block of $\mathbb{M}_q$ satisfies

$$M_{F \cup G} = \sum_{R \subseteq L} \mathbf{1}[F \subseteq R] \mathbf{1}[G \subseteq R] \widehat{M}_R.$$

Equivalently,

$$\mathbb{M}_q = (Z_{\text{edge}} \otimes I_{c_q}) \text{diag}_{R \subseteq L} (\widehat{M}_R) (Z_{\text{edge}} \otimes I_{c_q})^T. \quad (26)$$

Since the two zeta factors are invertible, rank is invariant under this congruence and equals the sum of the ranks of the diagonal blocks. Combining with [Theorem 7.9](#) proves (23).

For a one-edge boundary $L = \{e\}$, (26) is the concrete sanity check

$$\begin{pmatrix} M_\emptyset & M_e \\ M_e & M_e \end{pmatrix} \sim \begin{pmatrix} M_\emptyset - M_e & 0 \\ 0 & M_e \end{pmatrix}.$$

For the restricted connection matrix in which the boundary is required to be independent on each local side, only the single block $M_\emptyset$ is present. Its rank is c_q by (12) together with the realization-and-pinning lower bound. Finally, every $\widehat{M}_R$ is a $c_q \times c_q$ matrix, which gives (25). $\square$

The quotient of the vector space of q -boundaryed graphs by the radical of C_q is therefore a finite-dimensional commutative graph algebra of dimension r_q in the standard connection-matrix sense [24]. The case $q = 1$ is the explicit four-dimensional algebra of Theorem 5.4; higher q need not admit such a small presentation, but Theorem 7.10 determines their dimensions by a finite zero-forcing-specific kernel.

7.5. The first two nontrivial ranks. For $q = 1$ there are no boundary edges and $c_1 = 4$, so the general theory recovers

$$r_1 = 4.$$

For $q = 2$, the chain set has size $c_2 = 52$ and there is one possible boundary edge e . The Boolean transform has two sectors,

$$\widehat{M}_{\{e\}} = M_{\{e\}}, \quad \widehat{M}_\emptyset = M_\emptyset - M_{\{e\}}.$$

Proposition 7.11 (Machine-certified $q = 2$ sector ranks). *The exact finite matrices defined above satisfy*

$$\text{rank}_{\mathbb{Q}(x)} M_{\{e\}} = 44, \quad \text{rank}_{\mathbb{Q}(x)} (M_\emptyset - M_{\{e\}}) = 16. \quad (27)$$

Proof. This proposition is a finite exact computation, not an additional infinite structural argument. The supplementary certificate `separator_rank_certificate.json` fixes the chain-state order, canonical hashes of the integer coefficient matrices, the specialization $x = 2$, the prime 1,000,003, and explicit nonsingular minor indices. The independent verifier `verify_separator_rank_certificate.py` reconstructs all matrices from the definitions in this paper. Exact rational row reduction of the concatenated coefficient matrices gives upper ranks 44 and 16. At $x = 2$, exact elimination over $\mathbb{F}_{1,000,003}$ verifies nonsingular minors of orders 44 and 16, respectively, giving matching lower bounds because specialization cannot increase generic rank. No floating-point rank decision is used. $\square$

Corollary 7.12.

$$\boxed{r_2 = 60.} \quad (28)$$

Proof. Apply Theorem 7.10 and Proposition 7.11: $r_2 = 44 + 16 = 60$. $\square$

8. PENDANT-EDGE TRANSFER

The rooted product combines branches sharing a root. To recurse along a tree, we also need to move the root through an edge.

Let (H, u) be a rooted graph. Form a new graph $E(H, u)$ by adjoining a new vertex r adjacent only to u , and root the resulting graph at r . We call this operation *edge extension*. Let

$$(P_0, P_1, A_0, A_1)$$

be the profile of (H, u) , and let

$$(\hat{P}_0, \hat{P}_1, \hat{A}_0, \hat{A}_1)$$

be the profile of $E(H, u)$.

Theorem 8.1 (Edge-extension transfer). *The response polynomials satisfy*

$$\hat{P}_0 = A_1 + xA, \tag{29}$$

$$\hat{P}_1 = P_1 + xP, \tag{30}$$

$$\hat{A}_0 = P_0 + A_0, \tag{31}$$

$$\hat{A}_1 = 0. \tag{32}$$

Proof. Let $T \subseteq V(H) \setminus \{u\}$. A set counted by the new profile either contains u or does not.

Suppose first that u is initially blue; the corresponding sets are $T \cup \{u\}$ and contribute a factor x . With the new root r blue and forbidden from forcing, the evolution inside H is precisely the rooted process at u . Hence the new branch is passive whenever T is passive or active in (H, u) . If T is passive, then u does not spend its force inside H ; after H has completed, u can force an initially white r . These sets contribute xP to $\hat{P}_1$. If T is active, then completion of H consumes the unique force by u . With r initially white, the additional white neighbour r prevents the required internal force, and the source indicator is zero. These sets contribute xA to $\hat{P}_0$.

Now suppose that u is initially white. If T lies in P_1 , then vertices of $H - u$ force u and complete H without a force by u ; the vertex u may then force r . Hence P_1 contributes to $\hat{P}_1$. If T lies in A_1 , then vertices of $H - u$ force u , but the subsequent completion of H requires the unique force by u . The extra white neighbour r blocks that force when r is initially white. Hence these sets are passive with source indicator zero, contributing A_1 to $\hat{P}_0$.

Finally, if T lies in $P_0 \cup A_0$, then T does not force u . With r initially blue, the force $r \rightarrow u$ is both necessary and sufficient to place H in the corresponding successful rooted state. The new branch is therefore active and has source indicator zero. This contributes $P_0 + A_0$ to $\hat{A}_0$. An active state cannot source a pendant root, so $\hat{A}_1 = 0$. $\square$

It is useful to regard [Theorem 8.1](#) as a linear operator on the four-component profile. The rooted product is bilinear, while edge extension is linear; together they constitute a finite transfer calculus for trees.

8.1. Elementary checks. The following profiles provide useful checks on the conventions. The entries are ordered as (P_0, P_1, A_0, A_1) .

(H, r)	(P_0, P_1, A_0, A_1)	$\mathcal{Z}(H; x)$
K_1	$(1, 0, 0, 0)$	x
K_2 , root at an endpoint	$(0, x, 1, 0)$	$2x + x^2$
P_3 , root at an endpoint	$(x, x + x^2, 1, 0)$	$2x + 3x^2 + x^3$
P_3 , root at the middle	$(0, x^2, 0, 2x)$	$2x + 3x^2 + x^3$
C_3	$(2x, x^2, 0, 0)$	$3x^2 + x^3$
C_4	$(2x + x^2, 2x^2 + x^3, 0, 0)$	$4x^2 + 4x^3 + x^4$

For the star $K_{1,d}$ rooted at its centre, with $d \geq 2$, the profile is $(0, x^d, 0, dx^{d-1})$, which gives $\mathcal{Z}(K_{1,d}; x) = (1 + x)(x^d + dx^{d-1})$.

9. THE ZERO-FORCING POLYNOMIAL OF A TREE

Let T be a tree rooted at r . For a vertex v , let T_v denote the subtree induced by v and its descendants, rooted at v .

The one-vertex rooted graph has profile

$$(P_0, P_1, A_0, A_1) = (1, 0, 0, 0). \quad (33)$$

Indeed, the empty set is passive when the unique vertex is supplied as the root, but it cannot force an initially white root.

For each vertex v , recursively compute the profiles of its child subtrees. Apply edge extension to each child profile, thereby viewing the child branch as rooted at v , and combine the resulting branch profiles by the rooted product. At the global root, use (1).

Theorem 9.1 (Exact tree algorithm). *Let T be a tree on n vertices. The preceding dynamic programme computes the complete zero-forcing polynomial $\mathcal{Z}(T; x)$ exactly. With schoolbook polynomial convolution, it uses $O(n^2)$ integer arithmetic operations on polynomial coefficients.*

Let $M(\ell)$ denote the bit complexity of multiplying two ℓ -bit integers. A coarse implementation bound is then $O(n^2 M(n))$ bit operations. In particular, the schoolbook estimate $M(n) = O(n^2)$ gives $O(n^4)$. The output itself has $O(n^2)$ bits, and no optimized bit-complexity claim is intended.

Proof. Correctness follows by induction on $|V(T_v)|$. The base case is (33). Assume the child profiles at v are correct. By [Theorem 8.1](#), edge extension gives the exact response of each child branch at v . Since T_v is the rooted wedge of these branches, [Theorem 5.2](#) gives the exact profile at v . The induction closes, and [Theorem 4.2](#) reconstructs $\mathcal{Z}(T; x)$ at the global root.

For the arithmetic complexity, let the child-subtree sizes at a vertex v be $s_1, \dots, s_d$. Sequentially multiplying the dual-number polynomials associated

with the branches requires

$$O\left(\sum_{i=1}^d s_i + \sum_{1 \leq i < j \leq d} s_i s_j\right)$$

coefficient operations. The product $s_i s_j$ charges pairs of vertices lying in distinct child subtrees of v . Each unordered pair of vertices is charged at most once, namely at its lowest common ancestor. Summed over all v , these terms are therefore $O(n^2)$.

The linear edge-extension work is bounded by

$$\sum_{v \neq r} |V(T_v)| = O(n^2),$$

with the path giving the extremal order. Hence the total number of coefficient operations is $O(n^2)$.

Every coefficient is at most 2^n , and so has $O(n)$ bits. There are $O(n^2)$ scalar arithmetic operations on $O(n)$ -bit integers, which gives the stated bit bound. Finally, $\mathcal{Z}(T; x)$ has $n + 1$ coefficients of $O(n)$ bits each, so its binary output length is $O(n^2)$. $\square$

Corollary 9.2. *The zero-forcing polynomial of an n -vertex forest is computable in $O(n^2)$ integer arithmetic operations on polynomial coefficients.*

Proof. A set is zero forcing in a disjoint union if and only if its intersection with every connected component is zero forcing. The polynomial of a forest is the product of the polynomials of its tree components. Apply [Theorem 9.1](#) and multiply the component polynomials. $\square$

Remark 9.3. The algorithm returns the full distribution of zero-forcing sets. In particular, it returns the zero-forcing number $Z(T)$, the number $z(T; Z(T))$ of minimum zero-forcing sets, and every higher coefficient in the same computation.

10. TRANSFER THROUGH BLOCKS OF BOUNDED ORDER

We now pass from one-point unions to the block-cut tree. A *block* is a maximal connected subgraph having no cut vertex. Every bridge is treated as a two-vertex block. An isolated vertex is treated as a one-vertex block only when it is a connected component by itself. If a connected graph has a single block, that block is used as the root of the block-cut tree. Put

$$\beta(G) = \max\{|V(B)| : B \text{ is a block of } G\}.$$

The purpose of this section is to prove an exact algorithm whose exponential dependence is confined to $\beta(G)$.

10.1. Source and demand markers. Let (H, v) be a rooted graph and fix $S \subseteq V(H) \setminus \{v\}$. Its response state is one of

$$P_0, \quad P_1, \quad A_0, \quad A_1.$$

The second index records whether S sources an initially white root, while the letter records whether a supplied root has no remaining obligation or must expend its unique force inside H .

For $\omega \in \{P_0, P_1, A_0, A_1\}$ define

$$\text{src}(\omega) = \begin{cases} 1, & \omega \in \{P_1, A_1\}, \\ 0, & \omega \in \{P_0, A_0\}, \end{cases}$$

and

$$\text{dem}(\omega) = \begin{cases} 1, & \omega \in \{A_0, A_1\}, \\ 0, & \omega \in \{P_0, P_1\}. \end{cases}$$

A source marker colours the attachment vertex without external assistance. A demand marker is a new white pendant vertex which must be forced by the attachment vertex.

Lemma 10.1 (Local source–demand replacement). *Let (H, v) be rooted, let $S \subseteq V(H) \setminus \{v\}$ have response state ω , and let J satisfy $V(H) \cap V(J) = \{v\}$. Put $\Gamma = J \cup H$. Form the marker graph Γ_ω from J by adjoining a white pendant vertex d_v at v when $\text{dem}(\omega) = 1$. For an external initial set $Y \subseteq V(J)$, put*

$$Y_\omega = Y \cup (\{v\} \text{ if } \text{src}(\omega) = 1).$$

Then, for every forbidden set $Q \subseteq V(J) \setminus \{v\}$,

$$\text{cl}_\Gamma^{-Q}(Y \cup S) \cap V(J) = \text{cl}_{\Gamma_\omega}^{-Q}(Y_\omega) \cap V(J). \quad (34)$$

Moreover,

$$\text{cl}_\Gamma^{-Q}(Y \cup S) = V(\Gamma) \iff \text{cl}_{\Gamma_\omega}^{-Q}(Y_\omega) = V(\Gamma_\omega). \quad (35)$$

Consequently replacement preserves the four response states and failure at any distinguished external root $p \in V(J) \setminus \{v\}$.

Proof. Fix restricted sequences realizing $C^-(S)$ and $C^+(S)$. If ω is active, let u be the unique vertex of $N_H(v) \setminus C^+(S)$ and fix a restricted sequence which completes H after the force $v \rightarrow u$.

We prove the two inclusions in (34) by simulating arbitrary finite Q -restricted sequences. Internal saturation steps are allowed to be inserted during the simulation; they change no vertex of $J \setminus \{v\}$.

Marker-to-branch simulation. Begin in Γ by putting the H -part in its canonical internal configuration. If $v \in Y$, saturate directly from $S \cup \{v\}$ to $C^+(S)$. If $v \notin Y$ and $\text{src}(\omega) = 1$, saturate from S to $C^-(S) = C^+(S)$; this supplies v . If $v \notin Y$ and the source bit is zero, saturate from S to $C^-(S)$ and leave v white.

Now scan a finite legal sequence in Γ_ω . A force in J whose forcing vertex is different from v is copied whenever its target is still white, and is otherwise

omitted. Such a vertex has exactly the same neighbourhood in Γ and Γ_ω : no vertex of $J \setminus \{v\}$ is adjacent to $H \setminus \{v\}$ or to the demand leaf. Hence the copied force remains legal unless its target has already become blue. If a non-source branch first receives v from J , copy that force and then saturate *directly* from $S \cup \{v\}$ to $C^+(S)$, scanning the fixed saturation sequence from the current, possibly larger, blue set by [Lemma 2.4](#). This corrects the branch state without changing $J \setminus \{v\}$.

Suppose next that the marker sequence uses v as forcing vertex. If it forces a vertex of J , then no pending demand is present: while a demand leaf is white it is an additional white neighbour of v , and after v forces that leaf all neighbours of v are blue. Hence the corresponding branch is passive and has no white neighbour in H , so the same external force is legal in Γ unless its target is already blue. If the marker force is $v \rightarrow d_v$, then all neighbours of v in J are blue. In the active branch, u is the unique white neighbour of v in H ; replace $v \rightarrow d_v$ by $v \rightarrow u$ followed by the fixed restricted completion. Every vertex of J reached by the marker sequence is reached in Γ , and a completed marker graph expands to a completed original graph.

Branch-to-marker simulation. Start with an arbitrary finite Q -restricted sequence in Γ . First normalize its internal chronology. If $v \in Y$, prepend a saturation from $S \cup \{v\}$ to $C^+(S)$. If $v \notin Y$ and the source bit is one, prepend the fixed saturation from S to $C^-(S) = C^+(S)$, *regardless of which side coloured v first in the original sequence*. The source marker is therefore blue from the beginning. Scanning the original sequence from this larger blue set and omitting forces whose targets have already become blue is valid by [Lemma 2.4](#); in particular, a later external force to v may simply disappear. If the source bit is zero, prepend a saturation from S to $C^-(S)$. When an external force first colours v , insert a saturation directly from $S \cup \{v\}$ to $C^+(S)$ before continuing.

After this normalization, delete every force whose two endpoints lie in H , except for a force by v in an active branch. External forces with forcing vertex different from v remain legal in the marker graph for the same neighbourhood reason: their external neighbourhoods are identical, and a force is omitted only when its target is already blue. An external force by v can occur only for a passive branch: before an active branch is completed, its vertex u is an additional white neighbour of v , and after $v \rightarrow u$ the vertex v has already used its unique force. No pending marker demand obstructs a copied external force. Finally, contract the active force $v \rightarrow u$ to the marker force $v \rightarrow d_v$ and delete any remaining internal steps. In the total-completion case, the fixed internal completion may instead be appended after the corresponding marker force. The marker force is legal because legality of $v \rightarrow u$ in Γ implies that every neighbour of v in J is blue. This produces a legal marker sequence reaching every vertex of J reached by the original sequence; completion of Γ contracts to completion of Γ_ω .

Apply the first simulation to a maximal marker sequence and the second to a maximal original sequence. Confluence from Lemma 2.3 gives the two opposite inclusions in (34), and the completion statements in the two simulations give (35).

To compare response states at an external root p , first take $Q = \{p\}$ and $p \notin Y$. Equation (34) shows that p is reached in one p -restricted process exactly when it is reached in the other, so the source indicators agree even when the process later stalls. Next take $p \in Y$ and compare both $Q = \{p\}$ and $Q = \emptyset$. By (35) the p -restricted completion and the unrestricted completion agree in the two models. Theorem 4.2 therefore gives the same passive, active, or failure outcome, completing the proof. $\square$

10.2. The transfer table of a block. Let B be a block, let $r \in V(B)$ be its parent attachment, and let

$$U \subseteq V(B) \setminus \{r\}$$

be the child articulation vertices. For each $u \in U$, suppose that a descendant rooted graph has response profile

$$\mathbf{R}_u = (P_{u,0}, P_{u,1}, A_{u,0}, A_{u,1}).$$

For a map

$$\omega : U \longrightarrow \{P_0, P_1, A_0, A_1\},$$

let B^ω be obtained from B by adjoining one pendant demand marker at every u for which $\text{dem}(\omega(u)) = 1$. For $X \subseteq V(B) \setminus \{r\}$, take

$$X^\omega = X \cup \{u \in U : \text{src}(\omega(u)) = 1\}$$

as the initial blue set in B^ω , with r excluded. Let

$$\chi_{B,r}(X, \omega)$$

denote the response category of this concrete rooted configuration.

For $\gamma \in \{P_0, P_1, A_0, A_1\}$ define

$$T_\gamma(B, r; \mathbf{R}) = \sum_{\substack{X \subseteq V(B) \setminus \{r\} \\ \chi_{B,r}(X, \omega) = \gamma}} x^{|X|} \prod_{u \in U} R_{u, \omega(u)}, \quad (36)$$

where, for example, $R_{u, A_1} = A_{u,1}$.

Theorem 10.2 (Exact block transfer). *The rooted response profile of the union of B and all descendant graphs attached at the vertices of U , rooted at r , is*

$$(T_{P_0}(B, r; \mathbf{R}), T_{P_1}(B, r; \mathbf{R}), T_{A_0}(B, r; \mathbf{R}), T_{A_1}(B, r; \mathbf{R})).$$

Proof. Choose an initial subset X of the non-root vertices of B and, independently, an initial subset in every descendant graph. First discard descendant subsets that are blocked even when their attachment root is supplied. Such a subset cannot occur in a globally successful configuration: if the whole configuration completed, supply that attachment vertex at time zero and

delete the external step which first colours it; by monotone sequence simulation the descendant would then complete, a contradiction. Every remaining descendant subset lies in exactly one of the four successful rooted response states. By [Lemma 10.1](#), each such descendant graph may be replaced by its source and demand markers without changing the response category at r . The resulting category is therefore $\chi_{B,r}(X, \omega)$, and the generating weight of all choices yielding the fixed pair (X, ω) is

$$x^{|X|} \prod_{u \in U} R_{u, \omega(u)}.$$

The choices are disjoint and exhaustive. Notice that a source state at u and the independent choice $u \in X$ may lead to the same effective blue marker, but they are distinct original initial sets and carry weights differing by the factor x ; the explicit factor $x^{|X|}$ records this distinction. Summing according to the response category gives [\(36\)](#). $\square$

The theorem gives a finite kernel for every block. It is deliberately stated for an arbitrary block rather than only for a clique or a cycle.

10.3. Effective four-letter aggregation. The transfer table above keeps the choice of a block vertex separate from the four-state response of its descendant. For computation these choices can be aggregated without losing information. At a non-root child articulation u define

$$\lambda_u^{00} = P_{u,0}, \quad \lambda_u^{10} = xP_{u,0} + (1+x)P_{u,1}, \quad (37)$$

$$\lambda_u^{01} = A_{u,0}, \quad \lambda_u^{11} = xA_{u,0} + (1+x)A_{u,1}. \quad (38)$$

The first bit is the effective initial colour seen by the block and the second is the pending-demand bit. For example, a source state P_1 produces an effectively blue articulation whether or not the articulation itself was selected; the two disjoint choices have total weight $(1+x)P_{u,1}$. At a non-articulation block vertex the only possible letters are 00 and 10, with weights 1 and x .

Theorem 10.3 (Effective arbitrary-block kernel). *Let B have order m , parent root r , and q child articulations. For each assignment of one effective letter to every vertex of $V(B) \setminus \{r\}$, adjoin a demand leaf at every letter with second bit one and initially colour every vertex whose first bit is one. Classify the resulting concrete rooted marker graph as P_0, P_1, A_0, A_1 , or failure, and sum the product of its local letter weights over each successful class. The four sums are exactly the response profile of the decorated block.*

The number of concrete marker configurations is at most

$$4^q 2^{m-1-q} = 2^{m-1+q} \leq 4^{m-1}.$$

Proof. Expand each local weight in [\(37\)](#)–[\(38\)](#). Its summands are precisely the disjoint choices of descendant response state and selection/nonselection of the articulation that produce the same effective blue/demand pair. By [Lemma 10.1](#), all summands aggregated into a fixed letter have identical behaviour inside the parent block. Distributivity therefore turns the sum

in [Theorem 10.2](#) into the stated four-letter enumeration, with no loss or duplication of original initial sets. Ordinary block vertices contribute only the two letters 00 and 10. $\square$

10.4. Clique blocks of arbitrary order. For a clique block the effective configurations admit a closed transfer. Let $B = K_m$ have parent root r . At each non-root vertex v_i , use the letter weights (37)–(38), taking the singleton profile $(1, 0, 0, 0)$ when no descendant is attached. Put

$$B_i = \lambda_i^{10} + \lambda_i^{11}, \quad D_i = \lambda_i^{11}, \quad W_i = \lambda_i^{00} + \lambda_i^{01},$$

and define

$$\begin{aligned} B_* &= \prod_i B_i, & D_* &= \prod_i D_i, \\ W_* &= \sum_i W_i \prod_{j \neq i} B_j, & E_* &= \sum_i W_i \prod_{j \neq i} D_j. \end{aligned}$$

The term B_* counts configurations in which every non-root clique vertex is effectively blue, D_* the subfamily in which each such vertex also carries a demand, W_* configurations with exactly one effective white vertex, and E_* the latter subfamily in which every effective blue vertex carries a demand.

Theorem 10.4 (Clique transfer). *For a clique block of arbitrary order $m \geq 2$, the response profile at its parent root is*

$$\boxed{P_0 = D_* + W_* - E_*, \quad P_1 = B_* - D_*, \quad A_0 = E_*, \quad A_1 = 0.}$$

The four aggregates can be computed without division in one scan. Starting with $B_ = D_* = 1$ and $W_* = E_* = 0$, at vertex i update simultaneously*

$$\begin{aligned} B_* &\leftarrow B_* B_i, & D_* &\leftarrow D_* D_i, \\ W_* &\leftarrow W_* B_i + B_* W_i, & E_* &\leftarrow E_* D_i + D_* W_i, \end{aligned}$$

using the old values on the right.

Proof. A blue clique vertex carrying a white demand leaf cannot force inside the clique: until its demand is discharged the leaf is an additional white neighbour, and the demand can be discharged only after every clique neighbour is blue. Demand vertices are prohibited from supplying the unique force that colours a remaining white clique vertex.

With the root supplied, two or more effectively white non-root vertices block all forces into the clique, so only the cases counted by B_* and W_* can succeed. If all non-root vertices are effectively blue, the root-forbidden process completes. It has source bit one exactly when at least one effective blue vertex has no demand, yielding $P_1 = B_* - D_*$ and the all-demand remainder D_* in P_0 .

If exactly one non-root vertex is effectively white, then with an initially white root there are two white clique vertices and no source force can begin, so the source bit is zero. With the root supplied, a blue non-demand vertex can force the unique white clique vertex unless every effective blue vertex has

a demand. The former configurations have weight $W_* - E_*$ and are passive; in the latter, of weight E_* , the root must make the unique clique force and the state is active. Once the clique is blue all demands can be discharged. An active source state A_1 is therefore impossible. The scan recurrences are the standard division-free updates for the displayed products and one-exception sums. $\square$

10.5. The block-cut dynamic programme. Let G be connected. Root its block-cut tree at a leaf block and choose a non-articulation vertex of that block as the global root. Process nodes from the leaves towards the root. At an articulation node combine all child-block profiles by [Theorem 5.2](#). At a block node apply [Theorem 10.2](#). At the global root apply [Theorem 4.2](#). For a disconnected graph, multiply the polynomials of its connected components.

We use the following accounting convention. A profile returned across an attachment vertex represents only the vertices strictly below that attachment; the attachment vertex itself is excluded and is assigned to the parent block. At an articulation node, the represented sets of the child blocks are pairwise disjoint. At a block with parent attachment r , the represented set is the disjoint union of $V(B) \setminus \{r\}$ and the represented sets returned by its child articulations. This convention also explains why each initial selection factor x is introduced exactly once.

Lemma 10.5 (Pair charging on a decomposition tree). *Let a rooted decomposition tree have $O(n)$ nodes, with an absolute constant. At every node, partition the newly combined represented set into pairwise disjoint pieces of sizes $s_1, \dots, s_q$, and suppose the node returns its entire represented set as one piece to its parent. Suppose that the polynomial work at that node is*

$$O\left(\sum_i s_i + \sum_{i < j} s_i s_j\right).$$

If pieces are merged but never split on the path to the root, then the total work is $O(n^2)$ coefficient operations.

Proof. Charge each occurrence in a linear term to the ordered pair consisting of the represented vertex and the current decomposition node. Because the decomposition has $O(n)$ nodes, there are $O(n^2)$ such pairs. Charge a term $s_i s_j$ to unordered pairs of original vertices whose endpoints lie in the two corresponding pieces. Each unordered pair is charged at the unique lowest node at which its endpoints lie in distinct pieces. At every strict ancestor the two endpoints already belong to the same merged piece, so the pair is not charged again. There are at most $\binom{n}{2}$ unordered pairs. $\square$

Theorem 10.6 (Bounded-block algorithm). *Let G have n vertices and suppose every block has at most b vertices. The transfer tables can be constructed using*

$$O(4^b b^2 n)$$

elementary combinatorial operations. After this preprocessing, $\mathcal{Z}(G; x)$ is computable using

$$\boxed{O(4^b n^2)}$$

integer arithmetic operations on polynomial coefficients under schoolbook polynomial convolution. In particular, for fixed b the complete polynomial is computable in polynomial time.

Proof. Correctness follows by induction on the rooted block-cut tree from Theorems 4.2, 5.2 and 10.2.

For a block B with parent attachment r and q child articulations, there are at most

$$4^q 2^{|V(B)|-1-q} \leq 4^{b-1}$$

effective marker configurations by Theorem 10.3. Each response category is determined by a constant number of restricted-closure computations on a graph with at most $2b - 1$ vertices. With an adjacency matrix, or adjacency lists together with a queue of currently legal forcing vertices, each such computation uses $O(b^2)$ elementary graph operations. This gives the stated combinatorial preprocessing bound over $O(n)$ blocks.

For a fixed table entry, multiplying one chosen polynomial from child profiles of represented sizes $s_1, \dots, s_q$ has the cost in Lemma 10.5. Multiplying by the number of table entries gives the factor $O(4^b)$; the wedge operations at articulation nodes have the same form. The lemma yields $O(4^b n^2)$ coefficient operations. $\square$

Remark 10.7 (Arithmetic model and bit complexity). Every coefficient is at most 2^n and therefore has $O(n)$ bits. The theorem is a unit-cost count of integer arithmetic operations on polynomial coefficients, not a RAM-time bound. With schoolbook integer multiplication, the preceding implementation has the coarse bit bound $O(4^b n^4)$, in addition to the combinatorial preprocessing. No optimal bit-complexity claim is intended.

11. AN EXPLICIT CYCLE AUTOMATON

Finite automata have previously been associated with zero forcing: the zero-forcing automata of Shamsizadeh, Zahedi, Golmohamadian, and Abolpour, and the later k -forcing automata, use the vertices of the input graph as automaton states and encode a chosen forcing process and its recognized language [16, 17]. The construction below is of a different kind. It is a fixed-size fort-frontier recognizer whose input is a color-demand word around a cycle; its graph-independent five frontier states generate a 100-element Boolean transition monoid and a certified 32-state deterministic response quotient. The claim here is the specific finite frontier representation and its transfer consequences, not the introduction of automata into zero forcing.

A *cactus* is a graph in which any two cycles have at most one vertex in common; equivalently, every edge belongs to at most one cycle. Every block of a nontrivial connected cactus is either a bridge or a cycle; isolated vertices

are handled separately. The bounded-block algorithm is exponential in the order of a block, and the lengths of the cycle blocks need not be bounded. We now exploit their cyclic order. The result is a constant-state transfer algorithm which is independent of the cycle length.

11.1. Restricted forts. Let H be a graph and let $Q \subseteq V(H)$. A forcing process is *Q-restricted* if no vertex of Q is used as a forcing vertex.

Definition 11.1. A non-empty set $F \subseteq V(H)$ is a *Q-fort* if

$$|N_H(v) \cap F| \neq 1 \quad \text{for every } v \in V(H) \setminus (F \cup Q).$$

The usual forts are the $\emptyset$ -forts.

Lemma 11.2 (Restricted fort criterion). *Let $B \subseteq V(H)$ and let $t \in V(H)$.*

- (i) *The Q-restricted process from B colours all of H if and only if B intersects every Q-fort.*
- (ii) *The vertex t belongs to the Q-restricted closure of B if and only if B intersects every Q-fort containing t .*

Proof. Suppose that a Q-fort F is initially white. If a first force entered F , its forcing vertex would lie outside $F \cup Q$ and would have exactly one white neighbour in F , contrary to the definition. Hence no vertex of F is ever coloured.

Conversely, let W be the set of vertices left white at the terminal restricted closure. Every blue vertex outside Q has either no neighbour in W or at least two, since otherwise one further force would be available. Whenever $W \neq \emptyset$, it is a Q-fort disjoint from B . If a specified vertex t remains white, then $t \in W$. These observations prove both statements. $\square$

Consider a cycle block

$$C = v_0 v_1 \cdots v_{m-1} v_0, \quad v_0 = r,$$

where r is the parent attachment. After applying the replacement theorem of [Lemma 10.1](#), a non-root cycle vertex may carry a pending demand leaf.

Lemma 11.3 (Demand suppression). *A cycle vertex carrying a pending demand leaf never forces a cycle neighbour. Moreover, once all cycle vertices are blue, every pending demand can be completed. Consequently, for the purpose of deciding whether the marked cycle completes, a demand vertex may be treated as a vertex forbidden from forcing along the cycle.*

Proof. Suppose that a demand vertex v forces a cycle neighbour at some first such step. Its demand leaf must already be blue; otherwise the leaf and the target cycle neighbour would be two white neighbours of v . The leaf can only have been forced by v , and that earlier force required both cycle neighbours of v to be blue. This contradicts the choice of the later white target.

Conversely, if all cycle vertices are blue, then each demand vertex has its leaf as its unique white neighbour and may force it. The demands are independent, so all of them can be completed. $\square$

11.2. A five-state fort frontier. For $1 \leq i < m$, let b_i record effective initial blueness at v_i , and let d_i record a pending demand. For a candidate fort W , put $w_i = \mathbf{1}_{\{v_i \in W\}}$. At a non-root vertex the disjointness and fort conditions are

$$b_i w_i = 0, \quad w_i = 0 \text{ and } d_i = 0 \implies w_{i-1} = w_{i+1}. \quad (39)$$

Before processing the letter at v_i , a frontier state is

$$(w_{i-1}, w_i, h_i), \quad h_i = \bigvee_{j=0}^i w_j.$$

After choosing w_{i+1} and checking (39) at v_i , the state becomes

$$(w_i, w_{i+1}, h_i \vee w_{i+1}).$$

The only reachable triples are

$$q_0 = (0, 0, 0), \quad q_1 = (0, 0, 1), \quad q_2 = (0, 1, 1), \quad q_3 = (1, 0, 1), \quad q_4 = (1, 1, 1).$$

There is no sixth state because $h_i = 0$ forces both displayed fort bits to be zero.

For a letter (b, d) define the Boolean 5×5 matrix T_{bd} by

$$T_{bd}((a, c, h), (c, z, h \vee z)) = 1 \quad (40)$$

exactly when $bc = 0$ and $c = 0, d = 0 \implies a = z$. Row vectors act on the left, so the word $(b_1, d_1) \cdots (b_{m-1}, d_{m-1})$ is represented by

$$M = T_{b_1 d_1} \cdots T_{b_{m-1} d_{m-1}}.$$

An entry $M_{ab} = 1$ records a trace from initial frontier state q_a to final state q_b . The initial row encodes $(w_0, w_1, w_0 \vee w_1)$; after the last letter the final column encodes (w_{m-1}, w_0, h) , because closing the cycle sets the next bit equal to w_0 .

Let E_R, E_O, E_S denote, respectively, an obstruction with the root blue and forbidden, an obstruction with the blue root allowed to force, and an obstruction containing an initially white root. In the order $q_0, \dots, q_4$,

$$E_R(M) \iff \bigvee_{a \in \{0,2\}} \bigvee_{b \in \{1,3\}} M_{ab} = 1, \quad (41)$$

$$E_O(M) \iff M_{01} \vee M_{23} = 1, \quad (42)$$

$$E_S(M) \iff \bigvee_{a \in \{3,4\}} \bigvee_{b \in \{2,4\}} M_{ab} = 1. \quad (43)$$

For E_R the root bit is 0 and the final history bit enforces non-emptiness. For E_O the root lies outside the fort and is allowed to force, so its two cycle neighbours must have equal fort bits. Every ordinary fort counted by E_O is also a root-restricted fort counted by E_R ; hence $E_O \Rightarrow E_R$, and the response cases below are disjoint and exhaustive. For E_S the root bit is 1; non-emptiness is automatic and the fort condition at the root is omitted because the source test is carried out in the root-forbidden process, even when the root itself belongs to the fort.

Example 11.4 (An unmarked triangle). For a triangle with no selected non-root vertex and no demand, the word is 00 00. The product T_{00}^2 has, among others, the transition $q_0 \rightarrow q_0$, corresponding to the empty trace, and transitions $q_2 \rightarrow q_2, q_3, q_4$, corresponding to non-empty fort traces. In particular $E_O(T_{00}^2)$ holds, so the readout is failure: a supplied root alone does not zero force a triangle. If both non-root vertices are selected, the word is 10 10; the readout is P_1 , as the selected vertices force an initially white root and the cycle is already complete when the root is supplied.

Proposition 11.5 (Response readout). *The concrete response of the marked cycle word $\mathbf{a}$ is determined by $M = M(\mathbf{a})$ as follows:*

condition	response
$E_O(M)$	failure,
$\neg E_R(M), \neg E_S(M)$	P_1 ,
$\neg E_R(M), E_S(M)$	P_0 ,
$E_R(M), \neg E_O(M), \neg E_S(M)$	A_1 ,
$E_R(M), \neg E_O(M), E_S(M)$	A_0 .

Proof. By Lemmas 11.2 and 11.3, the plus process with the root forbidden completes precisely when $E_R(M)$ is false. This is the passive case. With the root allowed, the plus process completes precisely when $E_O(M)$ is false. By Theorem 4.2, the response is active exactly when $E_R(M)$ is true and $E_O(M)$ is false: the root-forbidden saturation fails, but one root force followed by another restricted saturation succeeds. If $E_O(M)$ is true, even the unrestricted plus process fails. Finally, the source bit is one precisely when there is no restricted fort containing the root, that is, when $E_S(M)$ is false. $\square$

The four matrices $T_{00}, T_{10}, T_{01}, T_{11}$ generate a finite Boolean transition monoid. The certificate described in Section A lists all 100 matrices in canonical breadth-first order and records their four right transitions and response outputs. Response equivalence identifies matrices having the same output after every continuation word. The supplied quotient map has 32 classes. For every pair of distinct quotient states, the certificate supplies a continuation word on which their outputs differ; hence the quotient is minimal as a deterministic response automaton. Minimality is not used for correctness or complexity—the unreduced 100-state monoid already suffices—but the smaller quotient is used in the reference implementation.

11.3. Weighted cycle transfer. Let $\mathbf{R}_i = (P_{i,0}, P_{i,1}, A_{i,0}, A_{i,1})$ be the descendant response profile at v_i ; at a non-articulation vertex use the singleton profile $(1, 0, 0, 0)$. Summing over whether v_i itself is selected gives the four letter weights

$$\lambda_i^{00} = P_{i,0}, \quad \lambda_i^{10} = xP_{i,0} + (1+x)P_{i,1}, \quad (44)$$

$$\lambda_i^{01} = A_{i,0}, \quad \lambda_i^{11} = xA_{i,0} + (1+x)A_{i,1}. \quad (45)$$

The first coordinate is the effective blue bit and the second is the demand bit. The aggregation is displayed explicitly below; “selected” contributes the factor x .

descendant state	v_i selected?	(b_i, d_i)	weight
P_0	0	00	$P_{i,0}$
P_0	1	10	$xP_{i,0}$
P_1	0	10	$P_{i,1}$
P_1	1	10	$xP_{i,1}$
A_0	0	01	$A_{i,0}$
A_0	1	11	$xA_{i,0}$
A_1	0	11	$A_{i,1}$
A_1	1	11	$xA_{i,1}$

The rows are disjoint at the level of original initial sets even when they produce the same effective letter, so no double counting occurs.

Let $\mathcal{A}$ be either the 100-state transition monoid or the 32-state quotient. Starting from the identity state e , define polynomial weights $f_i(s)$ by

$$f_0(e) = 1, \quad f_0(s) = 0 \quad (s \neq e),$$

and

$$f_i(t) = \sum_{\substack{s \in \mathcal{A}, (b,d) \in \{0,1\}^2 \\ s \cdot (b,d) = t}} f_{i-1}(s) \lambda_i^{bd}. \quad (46)$$

At the end, sum $f_{m-1}(s)$ over the states whose readout is respectively P_0, P_1, A_0, A_1 .

Theorem 11.6 (Explicit cycle transfer). *The four polynomials obtained from (46) are exactly the rooted response profile of the cycle block together with all descendant graphs attached at its non-root vertices.*

Proof. For every non-root cycle vertex, the four weights (44)–(45) form a disjoint and exhaustive partition of the choices of the block vertex and its descendant response state. Their product is the generating weight of the corresponding marked word. The deterministic transition in (46) computes its fort matrix, or its response-equivalence class. By Proposition 11.5, the terminal readout is exactly its rooted response category. Summing over all words proves the assertion. $\square$

11.4. The cactus algorithm. A cactus is processed on its rooted block-cut tree. A connected component consisting of one isolated vertex is a base case: its polynomial is x , and its rooted profile is $(1, 0, 0, 0)$. Every other connected component is processed as follows. The description is also pseudocode for the algorithm.

- (1) In each connected component choose a leaf block, and choose a non-articulation vertex r of that block as global root.
- (2) Process the block-cut tree in postorder.

- (3) At an articulation v , combine all already processed child-block profiles by the rooted product. The resulting profile represents the entire descendant graph attached at v ; the initial choice of v itself is not yet counted.
- (4) If a block is a bridge rv , apply the pendant-edge transfer to move the profile at v to the parent attachment r .
- (5) If a block is a cycle, orient it from its parent attachment $v_0 = r$ as $v_0v_1 \cdots v_{m-1}v_0$. Insert at each v_i , $i \geq 1$, the profile obtained in step (3), or the singleton profile when no child block is present. Apply the weighted cycle transfer. The factors x in (44)–(45) count the cycle vertex v_i exactly once.
- (6) Return the resulting block profile to the parent articulation. At the global root apply [Theorem 4.2](#); multiply component polynomials for a disconnected cactus.

Theorem 11.7 (Quadratic cactus algorithm). *The complete zero-forcing polynomial of an n -vertex cactus graph can be computed exactly with $O(n)$ graph-structural preprocessing and*

$$\boxed{O(n^2)}$$

integer arithmetic operations on polynomial coefficients under schoolbook polynomial convolution. The algorithm permits cycles of arbitrary length. A coarse schoolbook bit bound is $O(n^4)$.

Proof. Correctness follows by postorder induction from [Theorems 5.2, 8.1](#) and [11.6](#). Every non-root graph vertex is counted exactly once: at a bridge transfer if it is the child endpoint of a bridge block, or in one of the eight rows of the cycle-letter table if it lies in a cycle block. Articulation products combine descendant choices but introduce no new factor x .

The block-cut tree and cyclic orders are found in linear time. The automaton has constant size. At a cycle block, let s_i be one plus the number of vertices represented by the descendant profile inserted at the non-root cycle vertex v_i . The added one is v_i itself; the parent attachment v_0 is excluded. These pieces are pairwise disjoint under the convention preceding [Lemma 10.5](#). The scan costs

$$O\left(\sum_i s_i + \sum_{i < j} s_i s_j\right)$$

coefficient operations, up to an absolute automaton constant. Wedge products and bridge transfers have the same form. Apply [Lemma 10.5](#) to obtain $O(n^2)$ coefficient operations. The bit bound follows from the $O(n)$ -bit coefficient bound as in [Remark 10.7](#). $\square$

Theorem 11.8 (Clique–cycle block algorithm). *Let G be an n -vertex graph every block of which is either a clique or a cycle (with K_2 serving as a bridge*

block). Then the complete zero-forcing polynomial can be computed exactly after $O(n + |E(G)|)$ structural preprocessing using

$$\boxed{O(n^2)}$$

integer arithmetic operations on polynomial coefficients under schoolbook polynomial convolution. The block orders and cycle lengths are unrestricted. Consequently this gives an explicit quadratic coefficient-operation algorithm for every block graph as well as every cactus, and more generally for every block-cactus graph. The class has unbounded treewidth, since it contains K_m for every m , but it has bounded clique-width; see [23] and the discussion in Section 12.

Proof. Build the block-cut forest in $O(n + |E(G)|)$ time and process it in postorder. At articulation nodes use Theorem 5.2; at clique blocks use Theorem 10.4; and at non-clique cycle blocks use Theorem 11.6. (The edge K_2 may equivalently be handled by Theorem 8.1.) Each block transfer performs only a constant number of polynomial convolutions per represented piece. The represented interiors returned by distinct children are disjoint, so the charging argument of Lemma 10.5 gives $O(n^2)$ coefficient operations globally. For dense clique blocks the input itself may have $\Theta(n^2)$ edges, which is why the structural preprocessing is stated as $O(n + |E(G)|)$ rather than $O(n)$. $\square$

11.5. Independent random initial sets and vertex influence. The same transfer is naturally multivariate. Let every vertex v be selected independently with probability p_v , and write

$$Q_G(\mathbf{p}) = \Pr_{\mathbf{p}}(S \text{ is a zero-forcing set of } G).$$

Equivalently, one may regard $F_G(\mathbf{y}) = \sum_{S \text{ zero forcing}} \prod_{v \in S} y_v$ as a multivariate generating function represented by a division-free arithmetic circuit rather than by its exponentially large expanded coefficient table.

For a non-root block vertex v_i carrying descendant profile

$$(P_{i,0}, P_{i,1}, A_{i,0}, A_{i,1}),$$

the four effective source–demand letter weights become

$$\lambda_i^{00} = (1 - p_i)P_{i,0}, \quad \lambda_i^{10} = p_i P_{i,0} + P_{i,1}, \quad (47)$$

$$\lambda_i^{01} = (1 - p_i)A_{i,0}, \quad \lambda_i^{11} = p_i A_{i,0} + A_{i,1}. \quad (48)$$

At the final root r the readout is

$$Q_G(\mathbf{p}) = p_r(P_0 + A_0) + P_1 + A_1. \quad (49)$$

Corollary 11.9 (All vertex influences in linear scalar arithmetic). *On a cactus, or more generally on a graph whose blocks are cliques or cycles, with its block structure supplied, $Q_G(\mathbf{p})$ and all first partial derivatives $\partial Q_G / \partial p_v$*

can be evaluated using $O(n)$ scalar arithmetic operations. If the structure is not supplied, add $O(n + |E(G)|)$ preprocessing. Moreover,

$$\frac{\partial Q_G}{\partial p_v} = \Pr_{\mathbf{p}-v}(S \cup \{v\} \text{ succeeds}) - \Pr_{\mathbf{p}-v}(S \text{ succeeds}),$$

so by monotonicity this derivative is the probability that v is pivotal.

Proof. Replace every factor x and every unselected weight 1 in the clique/cycle transfer by the corresponding probabilities as in (47)–(48). The postorder algorithm then forms a division-free straight-line circuit with $O(n)$ addition and multiplication gates and terminal readout (49). The Baur–Strassen theorem gives all first partial derivatives of an arithmetic circuit with only constant-factor overhead [9]; equivalently, one may run the usual reverse-mode adjoint sweep on this particular circuit. All derivatives require $O(n)$ scalar operations. The displayed identity follows by conditioning on whether v is selected; multilinearity makes the derivative the difference of the two conditional success probabilities. Zero-forcing success is increasing under enlarging the initial set, so the difference is the pivotal probability. The contribution specific to the present setting is the explicit division-free $O(n)$ -gate heterogeneous success circuit, not the general reverse-mode differentiation principle. $\square$

Corollary 11.10 (Zero influence and fort-irrelevant vertices). *For any finite graph G , fix $v \in V(G)$ and suppose $0 < p_u < 1$ for every $u \neq v$. Then*

$$\begin{aligned} \frac{\partial Q_G}{\partial p_v} = 0 &\iff v \text{ belongs to no inclusion-minimal zero-forcing set,} \\ &\iff v \text{ belongs to no inclusion-minimal fort.} \end{aligned}$$

Proof. Conditioning on the membership of v in the product measure, or simply differentiating the multilinear polynomial Q_G , shows that the derivative is the probability, over subsets of $V(G) \setminus \{v\}$, that adding v changes failure into success. Full support on the other vertices makes this probability zero exactly when no subset $T \subseteq V(G) \setminus \{v\}$ is failed while $T \cup \{v\}$ is zero forcing. Such a witness exists exactly when v belongs to an inclusion-minimal zero-forcing set: minimize $T \cup \{v\}$ among its zero-forcing subsets containing v ; conversely removing v from a minimal zero-forcing set must fail. Jacob’s irrelevance theorem identifies the vertices lying in no minimal zero-forcing set with those lying in no minimal fort [15, Theorem 2.4]. The last equivalence is an application of that established characterization, not a new characterization of irrelevant vertices. $\square$

12. LOGICAL BOUNDED-WIDTH CONTEXT

The zero-forcing predicate is MSO_1 -definable: with vertex-set variables S and F , one says that F is a nonempty fort and that S meets every fort. Standard counting metatheorems therefore evaluate $\mathcal{Z}(G; x)$ on effectively bounded clique-width classes when an appropriate expression is supplied;

fixed-treewidth consequences follow from the usual MSO framework [20, 22]. These observations are background, not tractability claims of this paper.

They are relevant to the block-cactus application. Gurski records that the clique-width of a graph is at most two more than the maximum clique-width of a block [23, Sec. 3.6]. Since cliques and cycles have bounded clique-width, the class treated in Theorem 11.8 is already covered by the general metatheory. What Theorem 11.8 adds is an explicit small-state construction, a direct implementation, and an audited $O(n^2)$ coefficient-operation bound. Recent parameterized work on minimum zero-forcing sets and related variants concerns a different optimization or decision problem [10, 11].

13. COMPUTATIONAL VERIFICATION

The computations serve as checks or as finite certificates; none is used in place of an infinite proof. Independent subset enumeration was used wherever practical. The main checks are summarized below; the supplement contains the full logs, seeds, expected outputs, and commands.

Check	Cases or certificate size
Root normal form, edge transfer, wedges, articulation cases, and trees	34,625
Atlas and random bounded-block, cactus, and block-cactus comparisons	2,583
Marked cycle and clique words	174,756
Projected-closure replacement and decorated clique/block checks	69,966
Separator compatibility checks and realized chain states	8,284
Cycle certificate	100 matrices, 32 states, 496 witnesses
Two-terminal separator-rank certificate	ranks 52, 44, 16, 60

The separator verifier is independent of the block and cycle code. It builds the atom posets and chain states from the definitions, checks all 52 two-terminal chain realizations by direct prefort enumeration, samples the three-terminal construction, and reconstructs the $q = 2$ coefficient matrices. Exact rational elimination gives the symbolic upper ranks. Specialization at $x = 2$, followed by elimination modulo 1,000,003, gives matching lower ranks.

The cycle verifier reconstructs the four 5×5 frontier matrices, regenerates the 100-element Boolean transition monoid, checks the quotient to 32 response states, and verifies a distinguishing continuation for each of the 496 unordered pairs. The full audit also reconstructs the rank-four connection minor by direct subset enumeration and obtains $4x^6(x+1)^2$.

14. DISCUSSION

The separator theorem gives an explicit interface for every fixed boundary size. Its three ingredients are the prefert atoms, the compatibility rule at the boundary, and the chain compression. The realization theorem is what turns the last step from an upper-bound construction into an exact connection rank statement. The four-state rooted calculus is the case $q = 1$.

For independent boundaries the dimension is explicit: $r_q^\circ = c_q$. Boundary edges add Boolean Möbius sectors, leading to (23); already $c_2 = 52$ and $r_2 = 60$. A natural next problem is to understand the ranks of those sectors in closed form or asymptotically. It would also be useful to find presentations of the higher connection algebras comparable to the source/capacity presentation at $q = 1$, and to see which graph families force similarly small quotients of the universal interface.

The explicit one-terminal algorithms have a different purpose. The general separator algebra need not be an efficient implementation when q is large, whereas trees, cliques, and cycles collapse to very small concrete states. This is why the structural separator theory and the block algorithms are kept in the same paper.

We use existing machinery where it is appropriate. Connection matrices and graph algebras are classical [24, 25]; forts and fort transversals are standard in zero forcing [1, 26]; minimal-fort enumeration is treated in [6, 7, 8, 5]; and finite automata for forcing processes occur in [16, 17]. The claims made here concern the explicit separator interface, its chain quotient and realization, the resulting exact connection-rank formula, and the full-polynomial transfer systems. We do not claim priority for the general existence of connection matrices, forts, automata, bounded-width tractability, or automatic differentiation.

APPENDIX A. FINITE CERTIFICATES

Two finite claims in the paper are supplied with independent certificates. The file `cycle_automaton_certificate.json` lists the generated 5×5 Boolean matrices, the quotient map to the 32 response states, and a distinguishing continuation for each unordered pair of quotient states. The verifier rebuilds the monoid from the four generators in (40) before checking the listed data.

The file `separator_rank_certificate.json` records the exact coefficient-span and specialization data used for the $q = 2$ calculation. Its verifier reconstructs the finite kernels, checks the stated minors, and repeats the modular rank calculation. Both verifiers are independent of the main dynamic-programming implementations. The checksum manifest in the supplement identifies the files used for the reported calculations.

DATA AND CODE AVAILABILITY

The supplementary archive contains the source code, the two certificate files and their verifiers, the separator-realization audits, the rooted/tree/block/cycle implementations, pinned dependencies, expected outputs, and a checksum manifest. No external data set is used. The complete audit is run with `python run_all_audits.py`. The release environment used for the final audit was Python 3.13.5, NetworkX 3.6.1, and pdfTeX 1.40.26 (TeX Live 2025/dev). A permanent repository or archival DOI, once assigned, should point to the checksum-verified supplement distributed with this manuscript.

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